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Dissemination of antibiotic resistance genes (ARGs) via integrons in *Escherichia coli*: A risk to human health☆

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ABSTRACT

With the induction of various emerging environmental contaminants such as antibiotic resistance genes (ARGs), environment is considered as a key indicator for the spread of antimicrobial resistance (AMR). As such, the ARGs mediated environmental pollution raises a significant public health concern worldwide. Among various genetic mechanisms that are involved in the dissemination of ARGs, integrons play a vital role in the dissemination of ARGs. Integrons are mobile genetic elements that can capture and spread ARGs among environmental settings via transmissible plasmids and transposons. Most of the ARGs are found in Gram-negative bacteria and are primarily studied for their potential role in antibiotic resistance in clinical settings. As one of the most common microorganisms, *Escherichia coli* (*E. coli*) is widely studied as an indicator carrying drug-resistant genes, so this article aims to provide an in-depth study on the spread of ARGs via integrons associated with *E. coli* outside clinical settings and highlight their potential role as environmental contaminants. It also focuses on multiple but related aspects that do facilitate environmental pollution, i.e. ARGs from animal sources, water treatment plants situated at or near animal farms, agriculture fields, wild birds and animals. We believe that this updated study with summarized text, will facilitate the readers to understand the primary mechanisms as well as a variety of factors involved in the transmission and spread of ARGs among animals, humans, and the environment.

1. Introduction

Global challenge of the antimicrobial resistance (AMR) can be understood through an integration of the environment, animal health, agriculture field, and public health (Robinson et al., 2016). This emerging public and animal health-related threat is due to the circulation of resistant bacteria and antibiotic resistance genes (ARGs) among stated compartments (Woolhouse et al., 2015). ARGs which can be observed in various environments are considered as emerging environmental contaminants (Pruden et al., 2006; Ma et al., 2017). A survey showed ARGs can also be transferred and transformed in various environmental media such as ground water, water, soil, etc., and they can be integrated into environmental microorganisms through mobile genetic elements such as integrons, transposons and plasmids, and finally spread among bacteria through horizontal gene transfer (Wright, 2010). The mechanism of emergence and spread of ARGs is a complex process, which includes multiple factors such as: (1) heavy metals (Pal et al., 2017); (2) disinfectants (Horner et al., 2012); (3) the aggressive use of antimicrobials in the intensive agriculture and livestock farming (Thanner et al., 2016; Samanta and Bandyopadhyay, 2020); (4) wastewater treatment plant (WWTP) disposal (Park et al., 2018b); (5) aquaculture (Santos and Ramos, 2018); and (6) use of antibiotics either as growth promoters and/or disease prevention of animals (Ghosh et al., 2019; Theobald et al., 2019). The occurrence of ARGs has been frequently studied in gram-negative bacteria, for their

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possible role in AMR in clinical settings. However, the spread of ARGs and the emergence of multidrug resistant bacteria in outside clinical settings (natural environment) are also leading to public health issues. The pollution of ARGs to natural ecosystems does the evolution among the microbial population in the environment (Martinez, 2009). Out of various mechanisms involved in the spread of AMR, the genetic change/mutation and horizontal transmission of ARGs by bacteria are considered to be the main pathways leading to the risk of AMR (Davies, 1994; Martinez and Baquero, 2000). A continuous evolution and substitution in the plasmid-mediated ARGs are resulting in the emergence of novel enzymes and subsequent drug resistance. For instance, drugs like β-lactams antibiotics were previously effective against gram-negative bacteria, but a decrease in the efficacy of these antibiotics has been recorded with the rise of ARGs (Asai et al., 2011; Kogovšek et al., 2019). Remarkably, it was evidenced that the β-lactams resistance genes were associated with complex Class-1 integrons, which contained some β-lactamase-encoding genes such as $bla_{CTX\text{-}M\text{-}2}$, $bla_{OXA\text{-}1}$, and $bla_{NDM\text{-}1}$ (Guo et al., 2016; Le-Vo et al., 2019). Such a scenario results in failed prophylaxes and therapeutic in the human healthcare system and a negative impact on the socio-economic condition of the people. A sincere consideration in the subject matter is important, because if AMR left unattended, an estimate of 10 million human deaths per year has been predicted until 2050 (O'neill, 2014). Hence, it is not only need to recognize the harm of indiscriminate use of antibiotics but also to enforce compliance with regulations and preventive measures.

The meat industry is growing to meet the protein needs for an ever-escalating human population worldwide, at an affordable price (Henchion et al., 2014). There are two primary sources of animal protein, i.e. poultry and pig meat. An annual growth of 1.6% is estimated in the poultry industry, with an estimated production of 108.7 million tons of poultry meat (Harmse et al., 2016). On the other hand, swine meat covered 30% of meat consumption, worldwide (Soare and Chiurciu, 2017). Such an expansion in the industry has resulted to contaminate the environment through feces and farm waste materials, which leads to the mounting risk of infections to the general public (Headey and Hirvonen, 2016; Alam et al., 2019). Due to be influenced by several bacteria within the *Enterobacteriaceae*, several studies have indicated meat as a reservoir of ARGs and a potential source of the environmental contamination (Hasan et al., 2016; Dandachi et al., 2019; Poudel et al., 2019).

Escherichia coli (*E. coli*) is a common bacterium and widely distributed in natural environment, which can be isolated in many places, such as food-producing animals, farm waste materials, untreated water wastes, aquaculture, river bed, soil, farmland, etc. Various remarkable research works were under progress regarding relevant information of *E. coli*, such as clinical epidemiology from different sources, underlying phenomenon on the emergence, expansion and dissemination of drugs resistance. However, there is no systematic study exists, to summarize the transmission of drug resistance genes through integrons, mainly from *E. coli* of different reservoirs, which is harmful to the environmental safety. Here in this review, the published literatures were extracted from different search engines such as Science Direct, Pub Med, Google Scholar, CAB Abstracts, and Bing. The keywords were mainly including but not limited to ARGs, AMR, *E. coli*, environmental microorganisms, environmental pollutants/contaminants, human health, antibiotic resistance, integrons, Gene cassettes (GCs), antimicrobials, antibiotic residue, resistance exposure, WWTP, wild birds, wild animals, food-producing animals, and farm animals. Cited references were also used for further explorations. This article summarized the particular mechanisms involved in the spread of ARGs, and the prevalence of ARGs mediated by integrons among various environmental settings. Further, the negative implications of irrational uses of antibiotics and antibiotic residue as well as economic impact were examined. Moreover, the judicial uses of medicine, prevention management and control strategy to tackle the issue of AMR have been discussed. The main thrust of this article is to gather and organize so-far available information at a single platform, and build a bridge for relevant readers to understand that the dissemination of ARGs via integrons in *E. coli* from various reservoirs can pollute natural environment, which will risk human health.

2. Resistant *E. coli* and ARGs mediated environmental contamination

Enterobacteriaceae, gram-negative bacteria, are natural resident microbiome in the gastrointestinal tract (GIT) of all animals and humans. Among these, some *E. coli* with certain serotypes are highly pathogenic, i.e. extra intestinal pathogenic *E. coli* (ExPEC), which can cause urinary tract infections (UTI), septicemia and neonatal meningitis in humans (Bélanger et al., 2011; Messaili et al., 2019). Mainly, the bacterium spread in the environment via feces-contaminated water (Navab-Daneshmand et al., 2018), originating from soil (Ansari et al., 2008), farm waste disposal (Economou and Gousia, 2015), rivers (Dhawde et al., 2018) and marine water (Chern et al., 2009). Further, the bacterium can produce colisepticemia, airsaculitis, meningitis, perihepatitis, pericarditis, UTI and sepsis in human and animal population (Dho-Moulin and Fairbrother, 1999; Kayange et al., 2010; Asai et al., 2011).

Since 1990, the harmfulness of resistant bacteria, particularly *E. coli,* has dramatically increased because of its direct effects on the community healthcare system. Approximately, 70% infections of the GIT are believed to be originated from this particular bacterium (Laupland et al., 2008; Seiffert et al., 2013). Existence and infection of *E. coli* is further enhanced by the co-factors that include but not limited to age, gender, management status, and history of previously used medicines (Laupland et al., 2008; Theobald et al., 2019). Animal-originated food and contaminated environment in or around the commercial farms also took part in the development of multidrug resistance (MDR). Evidences have been recorded for the spread of resistant *E. coli* and linked mobile genetic traits from livestock to the human population (Seiffert et al., 2013; El-Demerdash et al., 2018). Isolated from farmers, dogs, flies and farms, the common bla_{NDM}-positive *E. coli* was identified by whole genome sequencing, which provided direct evidence for the circular spread of antibiotic resistant *E. coli* among different reservoirs and its ability to pollute the environment (Wang et al., 2017). In addition, the *E. coli* was able to affect public places including children's playgrounds, where sandpits were identified to harbor antibiotic resistant *E. coli* (Badura et al., 2014). It elaborates a variety of reservoirs and multiple pathways involved in the persistence, spread, and transmission of resistant *E. coli* strains among livestock, pig, poultry, waterfowl birds, domestic birds, companion animals, farm-worker, agriculture runoff, aqua environment, and healthy humans (Fig. 1).

Water serves multiple purposes in our routine life that includes but not limited to drinking, agricultural irrigation, everyday household use, boating, and swimming, etc. The *E. coli* can contaminate aquatic environment through feces-associated pollution (Fakhr et al., 2016). In this regard, there are multiple reports, where *E. coli* associated disease outbreaks were evidenced after the use of feces-contaminated water (Probert et al., 2017; Park et al., 2018a). The *E. coli* is one of the most common microorganisms in the natural environment and can be used as the indicator organism to examine developments and patterns of AMR, because they are

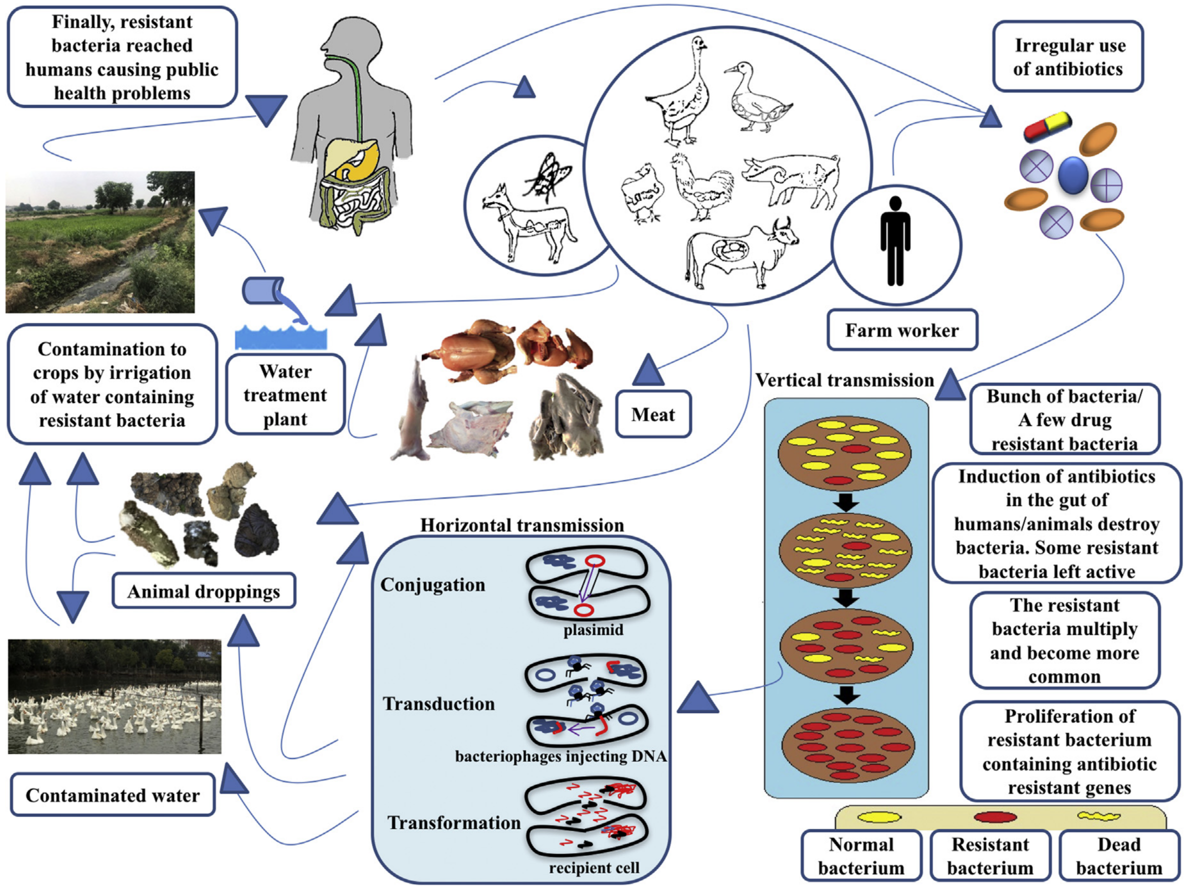

Fig. 1. Transmission routes illustration of antibiotic resistant *E. coli* strains in natural environment. Possible spread routes of antibiotic resistant *E. coli* showed by arrows among reservoirs such as ducks, geese, chickens, dogs, pigs, cattle, flies, WWTP, agricultural field, and humans. The horizontal and vertical transmission of ARGs among bacteria in the environment can cause public health issue finally.

widespread commensal in GIT of both humans and animals, as well as frequently available in environmental settings (World Health Organization, 2000; Navarro-Gonzalez et al., 2013; Chandler et al., 2018). Furthermore, *E. coli* can be cultured in the laboratory easily and economically, which can also capture ARGs by means of mobile genetic elements (Frye and Jackson, 2013; Devarajan et al., 2016; Partridge et al., 2018) In general, *E. coli* was an ideal pointer to study selective pressure forced by antimicrobials among food-producing animals (Harada and Asai, 2010; Zhang et al., 2015; Pouwels et al., 2019). Not surprisingly, the antibiotic resistant *E. coli* strains carrying different ARGs were widely studied as the possible environmental pollutant over the past decade, and their dissemination will have potential risks to human health (Harakeh et al., 2006; Su et al., 2012; Badura et al., 2014; Power et al., 2016; Ji et al., 2019; Nnadozie and Odume, 2019; Huijbers et al., 2020).

3. Integrons and its role in antimicrobial resistance and environmental contamination

Although integrons are as old as bacteria itself, the emergence and evolution within integrons have gained much attention during the last few decades, because of its relevance with antibiotic resistance (Kaushik et al., 2018). Further, environment plays significant role in the development of AMR (Larsson et al., 2018). The integrons are mobile genetic elements and consist of three main parts: they are the integron integrase gene (*intI*) (Messier and Roy, 2001), integron-associated recombination site (*attI*) along with GCs (Collis and Hall, 1995; Partridge et al., 2000), and the promoter (Pc) (Lévesque et al., 1994). This versatile structure of integrons enables them to accommodate and capture genes from more extensive genome diversity (Gillings et al., 2005). Basic integrons structure contained conserved regions [5′-conserved sequences (5′-CS) and 3′-conserved sequences (3′-CS)], which were separated by a variable region (VR). The truncated part 3′-CS consists of quaternary ammonium compound resistance gene (*qacE△1*) and sulfonamide resistance gene (*sul1*) (Zhang et al., 2019). Integrons can capture new genes through a reversible cassette-associated recombination process (Cameron et al., 1986). This results in subsequent evolution by the integration of new genetic material into bacteria via site-specific recombination (*attI*), and expression by the same promoter of host bacterial genome (Collis and Hall, 1995). Facilitated by integrons integrase, these GCs are integrated into *attI* and *attC*. The integrons are simple but compact DNA elements that have open reading frames and recombination sites (Fig. 2). The recombination site is known as *attC*, which is responsible for the production of important secondary stable structures for the identification of *intI* as well as recombination (Stokes et al., 1997). The complex Class-1 integrons could contain insertion sequence common region (IS*CR1*) element (Toleman et al., 2006) and more than one VR such as VR-1, VR-2, VR-3, etc. The VR-1 consists of complete 5′-CS and 3′-CS regions, but another exclusive VR such as VR-2 were found between IS*CR1* and another copy of the 3′-CS region (Quiroga et al., 2007). Interestingly, a recent study also showed the complex Class-1 integron can expand its structure and carry two 5′-CSs, three 3′-CSs and four VRs as well as multiple ARGs (Le-Vo et al., 2019). There

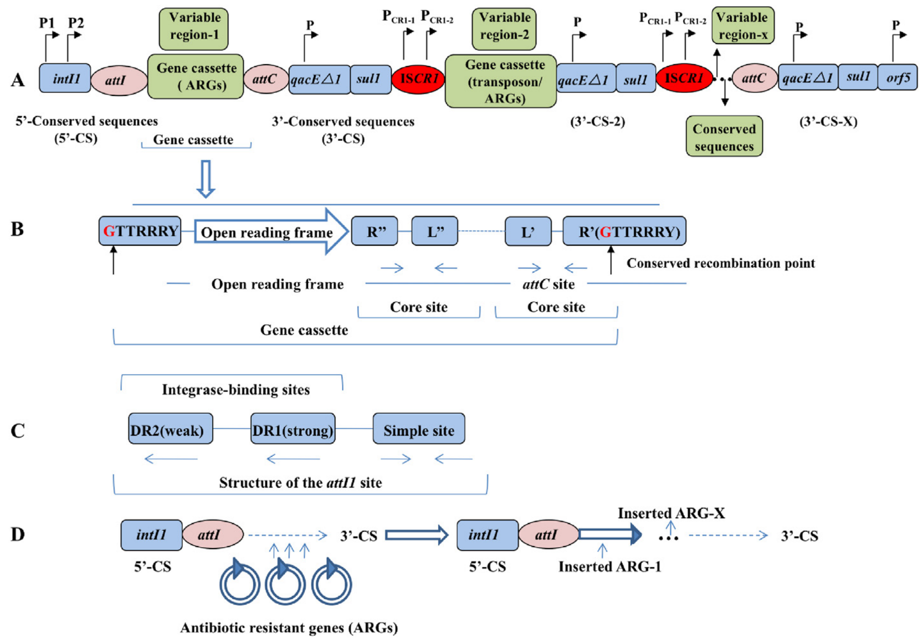

Fig. 2. Potential schematic illustration of Class-1 integrons/complex Class-1 integrons. (A) *P1* and *P2* are two promoters; *P1* is responsible for the transcription of gene cassettes while *P2* is generally inactive. The *intI1* represents integrase gene; *attI* stands for integration site; *qacE△1* means quaternary ammonium compound resistance gene; *sulI* stands for sulphonamide resistance gene; *orf 5* function is unknown and the *P* on *qacE△1* indicates internal promoters (Partridge et al., 2009; Xu et al., 2011). The complex Class-1 integron contained IS*CR1* followed by variable region (VR) and another copy of 3′-CS region (Quiroga et al., 2007) and other special complex Class-1 integron contained two IS*CR1*s along with Tn125-like transposon in its VR (Le-Vo et al., 2019). The P_{CR1-1} and P_{CR1-2} are two functional P_{OUT} promoters on IS*CR1*, which are responsible for the expression of downstream resistance genes (Lallement et al., 2018). (B) It is showing the structure of gene cassette along with integron recombination in linear formation; GTTRRY is the conserved position with start codon or recombination point, and the site of recombination followed by open reading frame; the R″, L″, L′, and R′ are integrase binding domains in *attC* (Gillings, 2014). (C) The *attI* structure contains direct repeats: DR1 (strong) and DR2 (weak) as well as a simple site for strong binding (Mazel, 2006; Deng et al., 2015). (D) Illustration of site-specific recombination for integration of resistant gene cassettes; the recombination occurred between the *attI* site and the *attC* site by incorporating ARGs; there may be more than one ARG cassette that can be inserted one by one at position of 3′-CS (Gillings, 2014; Deng et al., 2015).

were two functional P_{OUT} promoters (P_{CR1-1} and P_{CR1-2}) carried by IS*CR1* (Fig. 2), contributing to the expression of downstream resistence genes, and subsequently resulting in the presence of AMR in normal bacteria (Lallement et al., 2018). Moreover, the specific IS*CR1* played potential role in the dissemination of ARGs and the development of MDR (Le-Vo et al., 2019).

In general, integrons are characterized by the *intI* and *attI*, the reside sequence of *intI* determine the class of integrons. There are many types of integrons based on the *intI* gene, but three basic classes of integrons are well recognized i.e., Class-1 (*intI1*), Class-2 (*intI2*), and Class-3 (*intI3*) (Deng et al., 2015). Integrons are one of the most important cause of MDR with complex models via possession, expression, and distribution of resistant genes (Gillings, 2014). Each class of integrons behaved differently in the spread of AMR due to varied capability of capturing GCs (Escudero et al., 2015). The Class-1 integrons are more diverse and significant players in the spread of AMR (Gillings et al., 2008), while the Class-2 integrons have lower diversity and prevalence (Deng et al., 2015). A variety of appurtenance in Class-2 integrons made them successful for horizontal gene transfer and spread of ARGs in the environment (Gillings et al., 2008). The Class-3 integrons are rarely found and involved in the spread of MDR at lower rate (Kargar et al., 2014; Elizabeth et al., 2018). The recombination among a cassette associated 59-be and secondary sites was catalyzed by *IntI3*, and its frequency of occurrence was significantly lower than *IntI1* (Collis et al., 2002).

The IS*CR1* is belonging to an extended family of IS*91*-like elements which possibly mobilize neighboring DNA via rolling-circle replication (Toleman et al., 2006). The IS*CR1* is of 2.1 kb DNA genetic elements, exhibiting high mobility on the DNA portions (Song et al., 2010). The complex Class-1 integrons are carrying IS*CR1* element, which plays a vital role in the capturing and horizontal spread of ARGs (Song et al., 2010; Sun et al., 2012; Le-Vo et al., 2019). The documentation of the prevalence of IS*CR1* is increasing rapidly (Bae et al., 2008; Santos et al., 2011; Rui et al., 2018) with insertion sequences (IS) at the location of 3′-CS. Further, the incorporation of IS in Class-2 integrons, evolving them with diversification and further its presence on the chromosome, enabling them to transmit vertically (Alonso et al., 2018). Different classes of integrons carrying GCs were prevalent worldwide and the available data also showed a lower prevalence of Class-2 or 3 integrons (Table 1). Therefore, a continuous reporting of integrons is required for monitoring and minimizing MDR, in order to understand the spread of resistant gene spectrum, especially among emerging bacterial pathogens. In the subject matter, advanced techniques such as PCR, sequencing, and PCR cartography have enabled us to understand the emergence of novel mobile genetic elements (Gillings, 2014; Alonso et al., 2018; Zhang et al., 2019).

The survival of bacteria is mainly based upon their potential to adapt to the surrounding environment. There is a complex mechanism to understand the role of integrons in the evolutionary process of bacteria. Adaptability of bacteria sometimes depended upon the capturing ability of integrons, and transfer ability of the same plasmid, which can accelerate horizontal transmission of ARGs (Kaushik et al., 2018). This whole course resulted in the emergence of novel variants with the potential to express altered or recombined genes. Integrons-associated resistance development was occurred in multiple environment settings, which leaded to environmental contamination. These included rivers (Yang et al., 2018), soils (Chen et al., 2016), aquatic environment (Pérez-Etayo et al., 2018; Marathe et al., 2018), marine sediments (Chern et al., 2009), and waterfowls (Zhang et al., 2019), which were directly

Table 1
The prevalence of integrons classes and gene cassettes from *E. coli*.

Country	Bacteria	Prevalence of class	Sampling (Year)	Sampling (Source)	Variable region/Gene cassettes	Reference
China	*E. coli*	Class 1 and 2	NA	Giant panda	*dfrA17 + aadA5, aadA2, dfrA12 + aadA2i, dfrA1 + aadA1, dfrA1, aadA1*	Zhu et al. (2020)
India	*E. coli*	Class 2 and 3	NA	Poultry, Cattle, Pig, Goat Environment,	NA	Tewari et al. (2019)
China	*E. coli*	Class 1 and 2	2018	Water fowls	*dfrA1-orfC, aadA2, aadA1, dfrA1-adA1, dfrA1-orfC-aadA1, dfrA1-sat2-aadA1*	Zhang et al. (2019)
USA	*E. coli*	Class 1	2015–16	Water	NA	Cho et al. (2019)
Algeria	*E. coli*	Class 1	NA	Broiler chicken	NA	Messaili et al. (2019)
Egypt	*E. coli*	Class 1 and 2	NA	Poultry	*dfrA12*-orfF-*aadA27, aadA23, dfrA15, dfrA1-sat2-aadA30, sat2-aadA1, catB2, sat2*	El-Demerdash et al. (2018)
Pakistan	*E. coli*	Class 1, 2 and 3	NA	Livestock, Poultry	NA	Rahman et al. (2018)
Argentina	*E. coli*	Class 1	NA	Biopurification system	*aadA1b, aadA2, aadA11cΔ, dfrB3-aadA1di-catB2-aadA6k*	Martini et al. (2018)
South Korea	*E. coli*	Class 1	2016–17	WWTP	*aadA2, aadA22, aadA12, dfrA15*	Park et al. (2018b)
Quebec Canada	*E. coli*	Class 1	2009–2013	Poultry	*aadA, aac(3)-VI*	Chalmers et al. (2017)
China	*E. coli*	Class 1 and 2	2015–2016	Yaks, Piglets, Chickens	*dfrA1, dfr12, aadA1, aadA2, sat 1, orfF*	Rehman et al. (2017)
Egypt	*E. coli*	Class 1	NA	Poultry	*dfrA1, dfrA15, dfrA1*-orf	Moawad et al. (2017)
Nigeria	*E. coli*	Class 1	2009–2014	Poultry, Restaurants	*aadA2*-orfF-*dfrA12*	Ojo et al. (2016)
Egypt	*E. coli*	Class 1 and 2	2014–2015	Poultry	*dfrA5, dfrA7, dfrA12, aadA2, aadA5, aadA23, dfrA1, sat2, aadA1, sat2*	Awad et al. (2016)
Estonia	*E. coli*	Class 1 and 2	2013	Cattle slurry, Soil	*sul1*	Nõlvak et al. (2016)
Antarctica	*E. coli*	Class 1	2009–2010	Seawater, Marine sediments	*aadA1, aadA2, aadA4, aadA5-dfrA17*	Power et al. (2016)
Poland	*E. coli*	Class 1	2014	WWTP, River water	*sul1, sul2*	Koczura et al. (2016)
Germany	*E. coli*	Class 1 and 2	2012	Pig manure	*sul1, sul2, tetA, tetX, tetM, qacEΔ1*	Wolters et al. (2016)
Norway	*E. coli*	Class 1 and 2	2003	Pork meat, Human	*dfrA1-aadA1, aadA1, dfrA12*-orfF-*aadA2, oxa-30-aadA1, dfrA1-sat 1-aadA1, dfrA17-aadA5*	Sunde et al. (2015)
Iran	*E. coli*	Class 1 and 2	2013	Aqua fish field	*tetA, qnrA, sul1, sul2, qnrA, cmlA, aac(3)IIa*	Tajbakhsh et al. (2015)
India	*E. coli*	Class 1	2013–2014	Poultry, Cattle	*aadA2, dfrA12, sul1, qnrB*	Kar et al. (2015)
Brazil	*E. coli*	Class 1	2011–2013	Poultry	*dfr, aadA*	Casella et al. (2015)
Italy	*E. coli*	Class 1 and 2	2009–2010	Italian turkeys	*dfrA, aadA, sat2*	Piccirillo et al. (2014)
Taiwan	*E. coli*	Class 1	2011–2012	Well water, Waste water, River water, Soil, Sludge	*sul1*	Hsu et al. (2014)
Nigeria	*E. coli*	Class 1 and 2	2008	Poultry	*sul1, sul2, sul3, aadA, strA, strB, catA 1, cmlA1*	Adelowo et al. (2014)
Spain	*E. coli*	Class 1	NA	Birds	*dfrA, aadA*	Sacristán et al. (2014)
Germany	*E. coli*	Class 1	NA	Poultry	*sul1*	Schwaiger et al. (2013)
Spain	*E. coli*	Class 1 and 2	2008–2009	Pig	NA	Marchant and Moreno (2013)
Azores islands	*E. coli*	Class 1	2010	Wild birds, Game bird	NA	Santos et al. (2013).
South Korea	*E. coli*	Class 1 and 2	2011	Poultry	*aadA1 + dfrA1, dfrA1 + sat2 + aadA1, sul1, sul2, tetA, tetB, aadA5 + dfrA7, dfrA12 + aadA2,* *aadA2 + aadA1 + dfrA12, aadA5 + aadA2/dfrA7* *aadA5 + dfrA7, dfrA12 + aadA2, aadA2 + aadA1 + dfrA12, aadA5 + aadA2/dfrA7*	Dessie et al. (2013)
Mexico	*E. coli*	Class 1	2002–2003	Chicken litter	*aacA4-catB3-dfrA1, aadA1, dfrA1*	Ponce-Rivas et al. (2012)
China	*E. coli*	Class 1 and 2	2008–2009	River catchment	*aac(6′)-Ib-cr-aar-3-dfrA27-aadA16, aacA4-catB3-dfrA1, aadA2-lnuF*	Su et al. (2012)
Portugal	*E. coli*	Class 1 and 2	2007	Chicken products	*dfrA1 + aadA1, dfrA1 + sat2 + aadA1, tetA, tetB, aadA, sul1, sul2, sul3*	Silva et al. (2012)
Tunisia	*E. coli*	Class 1 and 2	NA	Poultry meat	*estX-psp-aadA2-cmlA1-aadA1-qacH*-IS440-*sul3-orf1-mef(B)*Δ-IS26*, dfrA1-sat-aadA1, estX* *-psp-aadA2-cmlA1-aadA1-qacH*-IS440-*sul3-orf1-mef(B)*Δ-IS26 , *dfrA1-aadA1, dfrA17-aadA5,* *dfrA12-orfF-aadA2, dfrA14*	Soufi et al. (2011)
Czech Republic	*E. coli*	Class 1	2005–2006	Rooks	*aadA, sul1, tetA*	Literák et al. (2007)

Wastewater treatment plant at livestock farms (WWTP).
Not Available (NA).

or indirectly linked to the dissemination of ARGs and thus resulting in the contamination of natural environment. In this study, we will briefly discuss how these factors (mentioned above) affect and spread ARGs among various environmental settings.

4. Occurrence of integrons in food chain and its role in the dissemination of ARGs

Food is the primary commodity of our daily life. A significant proportion of food, particularly the protein, comes from animals such as pig and poultry (Henchion et al., 2014). Recently, reports have been shown the evidence for the transmission of resistance genes from animal meat to humans (Sun et al., 2019; He et al., 2019). Obviously, the poultry and pork meat are essential parts of human food, since they occupy the significant proportion of protein demand. The consumption of chicken and pork meat per capita per year in the worldwide was recorded as 14.2 kg and 12.3 kg, respectively. In addition, USA is the most significant poultry meat consumer, i.e. 49.7 kg per person per year, while Europe stands second position with 23.6 kg per person per year (OECD, 2020). Surprisingly, about 80% of rising pigs received antibiotics as a feed additive (Li, 2017). The consumed antibiotics were poorly utilized and excreted via urine and feces as metabolites (Lienert et al., 2007; Qiu et al., 2016). The drugs as well as their metabolites of drugs were putting positive pressure on bacterial population (Heuer et al., 2008; Xiong et al., 2018b), making animals a potential source of antibiotics residues.

China is the largest producer of white-leg shrimp, providing aqua-origin protein to humans. Owing to its low price, the sulfonamides are extensively used in aquaculture to control and prevent infections. This may be the reason that from markets in Zhejiang province of China, a co-occurrence of Class-1 integrons with ARGs corresponding to sulfonamides resistance was revealed in *Penaeus vannamei* (white-leg shrimp) and pork meat (Jiang et al., 2019). Similarly, a prevalence of Class-1 and 2 integrons have been reported from pigs in Spain (Marchant and Moreno, 2013). The spread of *E. coli* with ARGs in food products is of great concern to public health. Integrons are one of the main contributors involved in the spread of AMR, with the help of ARGs, consequently polluting the environment. The prevalence of integrons as well as GCs associated with *E. coli* is summarized in Table 1 and Supplementary Fig. S1.

The spread of ARGs conferring resistance to tetracyclines, sulphonamides, and aminoglycosides in the food chain have been studied in raw meat (pork, chicken breasts, and beef), aquatic products (shrimps, fishes, and shellfishes), fresh vegetables (Chinese cabbages, lettuces, carrots, watercress, and tomatoes), and fresh fruits (pears, oranges, cherry tomatoes, tangerines). These ARGs associated with Class-1 integrons were more abundantly found in agricultural products as compared to aquatic food in China (Xiong et al., 2019). Further, Class-1, 2 and 3 integrons have been characterized from organic as well as conventional system fruits and vegetables (Jones-Dias et al., 2016). The investigations from the livestock sector identified Class-1 and 2 integrons along with six GCs inside variable regions. A frequent occurrence of GCs [dihydrofolate reductase genes (*dfrA1*) and aminoglycoside adenyltransferase genes (*aadA1*)] has been observed with the possibility of lateral gene transfer among food products (Wu et al., 2015). During 2007–2014, another study from Eastern China revealed the prevalence of Class-1 integrons conferring resistance against a range of antibiotics i.e. 97.5% against the tetracyclines (Dou et al., 2016). Moreover, from other parts of the globe including Mexico, Spain and Tunisia, meat products (pork and poultry) and fecal samples of pets (dogs and turtles), humans, wild animals (gull, genet, stork, and roe deer) were found as potential source of Class-2 integrons (Alonso et al., 2018). The Class-1, 2 and 3 integrons were reported from Pakistan along with India between livestock and poultry (Rahman et al., 2018; Tewari et al., 2019), indicating a threat to the emergence of novel classes of integrons in Asia.

There was some evidence that ARGs could be transferred to food chains, which would risk human health. For instance, originating from poultry meat in Egypt, the Class-1 integrons GCs (*dfrA12-orfF-aadA27*, *aadA23*, *dfrA15*) conferring resistance to aminoglycosides (streptomycin, spectinomycin) and trimethoprim were recorded. The Class-2 integrons GCs (*dfrA1-sat2-aadA30*, *sat2-aadA1*, *catB2*, *sat2*) conferring resistance to aminoglycosides, trimethoprim, streptothricin and chloramphenicol have been reported from diseased poultry and humans (El-Demerdash et al., 2018). From Nigeria (2009–2014), the Class-1 integrons were reported from samples of commercial poultry farms, household poultry, meat shops, live-bird markets, slaughter houses and restaurants (Ojo et al., 2016). While doing a comprehensive study on 29 chicken products in Portugal, Class-1 and 2 integrons were identified with ARGs conferring resistance to tetracyclines, streptomycin, and trimethoprim-sulfamethoxazole. Altogether, it suggested that ARGs from food products could flow to humans eventually (Silva et al., 2012). Originating from Tunisia, similar findings have also been reported (Soufi et al., 2011). In Canada, it was reported that the complex Class-1 integrons conferred resistance to gentamicin from poultry meat (Chalmers et al., 2017). Further, from chicken shops of Brazil, the complex Class-1 integrons carrying IS*CR1* as well as GCs (*aadA* and *dfr*) were also reported (Casella et al., 2015).

Keeping in view the significance of understanding AMR, the monitoring programs is on the way for last 15 years at the state level in Norway. While conducting AMR monitoring programs at Norwegian state, GCs for Class-1 and 2 integrons were reported from pork meat and human blood stream infections. Further, it was found that incF plasmids were used as vehicles in a horizontal spread of Class-1 integrons in the food chain (Sunde et al., 2015). Such an occurrence of similar but plasmid oriented ARGs indicated the potential transfer of ARGs from pork meat to humans. While conducting a comprehensive study on diverse livestock dairy farms in China, a spread of Class-1 integrons in the environment via milk has been evidenced (Ali et al., 2016).

An oro-fecal route of transmission for Class-1 integrons has been reported in ExPEC strains from Algeria (Messaili et al., 2019). The Class-1 integrons with sulphonamides resistance have been reported from multiple poultry flocks in Germany, along with clonal dissemination of ARGs to the environment (Schwaiger et al., 2013). Avian pathogenic *E. coli* (APEC) with Class-1 and 2 integrons conferring resistance to trimethoprim, streptomycin/spectinomycin, and streptothricin have been evidenced from lungs, spleen, liver, and air sacs of birds (Awad et al., 2016). The APEC originating from turkey fecal samples and commercial meat revealed a significantly high prevalence of Class-1 and 2 integrons. The GCs responsible for resistance against trimethoprim, streptomycin/spectinomycin, and streptothricinin were reported from Italy. Also, the variety of samples used in the study were taken from the brain, pericardium, lungs, tibiotarsal and coxo-femoral joints, liver, and spleen (Piccirillo et al., 2014). The prevalence of Class-1 integrons with ARGs conferring resistance to sulphonamides and tetracyclines was reported from fecal samples of ducks, chickens, swine, dairy, and beef cattle in South Korea (Unno et al., 2010).

5. Occurrence of ARGs in the aquatic environment

The aquatic environment got easily contaminated by waterfowl droppings, livestock, agricultural and WWTP outflows (Liao et al., 2018; Wambugu et al., 2018). The aquatic environment served as an essential habitat for *E. coli*, where potential ARGs were known to be transferred from resistant to non-resistant bacteria through plasmid-mediated integrons (Pazda et al., 2019). The mechanism for the spread of MDR bacteria from the human and animal population to the aquatic environment may be driven by WWTP, which discharged treated wastewater to main water reservoirs (Koczura et al., 2012; Lekunberri et al., 2017). Moreover, the role of phages in the spread of ARGs cannot be ignored (Lekunberri et al., 2017). The bacteriophages are another emerging source and vector for horizontal transfer of ARGs within aquatic environment. For instance, six clinically important ARGs and Class-1 integrons were revealed from water of Funan River in Chengdu, China (Yang et al., 2018). As such, the spread of ARGs may be involved with mobile genetic elements or bacteriophages in the aquatic river sediment settings, where significant seasonal variations in occurrence and abundance of ARGs were recorded (Calero-Cáceres et al., 2017). Several other factors contributed to the contamination of the aquatic environment by providing the source of ARGs, which included waste disposal from animal houses (Park et al., 2018b), storm water (Sidhu et al., 2014; Garner et al., 2017), industrial and hospital effluents (Devarajan et al., 2015), sewerage overflow, and animal feces (He et al., 2016; Sidhu et al., 2017). Further, the water quality in swine farm WWTP and microbial community were reported to be able to affect the prevalence of ARGs (Sui et al., 2019). Hence, the harmfulness of ARGs cannot be ignored in the aquatic environment, because various ARGs and antibiotics were prevalent in wastewaters and swine manure, which will risk water quality and food safety (He et al., 2016). According to the survey, the prevalence of resistant *Enterobacteriaceae* was highest in the river systems, just right after lakes, or followed by dams and ponds (Nnadozie and Odume, 2019). In addition, gut and skin microbiota from human can also contaminate water environment by disseminating ARGs (Baron et al., 2018; Zhou et al., 2018; Lee et al., 2020). It's also very worrying that natural dilution and degradation cannot eliminate ARGs from the aquatic environment (Nnadozie and Odume, 2019).

Different classes of integrons were prevalent and reported from aqua environment. For instance, from Taiwan area, the swine farm WWTP and downstream environment disposals were recorded as a source of resistant *E. coli* with Class-1 integrons, where GC (*sul1*) conferred resistance to sulfonamides (Hsu et al., 2014). A varying percentage of Class-1 (67.39%) and Class-2 (17.39%) integrons were reported from the irrigated water, which had a known history of contamination with animal and human feces in the Philippines (Paraoan et al., 2017). The prevalence of Class-1 integrons was observed in surface water in Brazil, where ARGs were conferring resistance to trimethoprim, streptomycin/spectinomycin, and quaternary ammonium compounds (Canal et al., 2016). Screening of the integrase genes from WWTP discharge, river water, and marine outfalls in Poland revealed the presence of Class-1 and 2 integrons (Koczura et al., 2012; Kotlarska et al., 2015; Koczura et al., 2016). The Class-1 and 2 integrons have also been evidenced in aquaculture, where ARGs were conferring resistance to aminoglycosides and trimethoprim (Lin et al., 2016). Moreover, it was revealed that the Class-1 integrons might have the potential to enter the human food chain (Gillings et al., 2008). A severe impact of human anthropogenic actions such as sewage disposal was observed in Antarctic marine ecosystems, where Class-1 integrons were found to contain multiple ARGs cassettes in Antarctic seals (Stark et al., 2016; Power et al., 2016).

6. Prevalence and factors affecting the spread of ARGs in agriculture

The frequent use of animal manure as organic fertilizer was a common practice to enhance the production of the land for an extended period. The manure from livestock, poultry, and swine was a significant source of ARGs (Han et al., 2018). The spread of ARGs with such a practice has been well documented (Heuer et al., 2011; Zhou et al., 2019a; Amador et al., 2019; Urra et al., 2019; Li et al., 2020c; Rutgersson et al., 2020). Some studies also showed that 30%–90% of administered antibiotics were excreted by animals and humans in the environment via feces or urination (Sarmah et al., 2006; Subbiah et al., 2012; Gillings, 2013). Hence, an increased use of antibiotics in the livestock augmented excretion of antibiotic or residues in the farm environment. Further, the antibiotics were excreted to the environment through animal manure and human waste material fixed with soil particles of agricultural land (Silva et al., 2017). The settled bacterial communities in soil could transfer ARGs to birds as well as other animals, making them as vectors, reservoirs, or bio-indicators for the dissemination of ARGs in the natural environment. The soil biota consisted of various enriched diversified intrinsic resistome, while human anthropogenic activities were playing an essential role in rising AMR worldwide (Zhu et al., 2019). A maximum occurrence was reported in pig manure followed by poultry and livestock (Duan et al., 2019). The use of sewage along with chicken manure was another cause of abundant ARGs in soil (Chen et al., 2016).

Recently, a large number of ARGs and Class-1 integrons carried by phages isolated from chicken manure have been reported from China (Yang et al., 2020). A survey from pig manure in Germany was conducted to evaluate the linkage between natural organic fertilizer and the spread of ARGs. The sulphonamides along with quaternary ammonium compounds were identified in the surrounding environment (Wolters et al., 2016). The application of organic manure to soil increased the ARGs for sulfonamides, aminoglycosides, and tetracyclines (Liu et al., 2017). The pig feces were significant reservoir of ARGs carried by bacteriophages in farm environment (Wang et al., 2018b). The samples from pig farms and multiple environmental settings harbored ARGs with abundance of *mcr-1*, pointing out potential public health risk (Ji et al., 2019). An occurrence of Class-1 and 2 integrons have been reported from Estonia, the similar ARGs were identified from agricultural grassland soil and cattle slurry (Nõlvak et al., 2016). The Class-1 integrons were prevalent in soil receiving animal manure (Heuer et al., 2012; Jechalke et al., 2014). The on-farm biopurification system was founded as a tank for Class-1 integrons with novel GCs conferring resistance to chloramphenicol, streptomycin and spectinomycin (Martini et al., 2018).

There are multiple factors involved in the spread of ARGs in the environment. In the evaluation of the effect of anaerobic digestion on the degradation of ARGs, it was found that there was no efficient removal of ARGs from swine manure via normal anaerobic digestion. It was concluded that the *tetG* gene persisted in the agricultural environment for a more extended period and thus facilitating the spread of ARGs (Cheng et al., 2016). However, full-scale advanced anaerobic digester treatment could reduce ARGs of sulphonamides by 5%–10% (Wallace et al., 2018). The increased pH and temperature during the storage of cattle slurry significantly reduced the ARGs load in animal manure (Li et al., 2020a). Other factors involved in the reduction of ARGs are including alkaline, and thermophilic conditions of slurry (Li et al., 2020b). It was observed that composting manure, as well as 120-days incubation significantly reduced the ARGs abundance in amended soils, which will be helpful to solve the menace of soil resistome (Chen et al., 2019).

7. Spread of ARGs via wild birds and animals

The wild birds were emerging potential reservoirs of resistant bacteria and involved in the spread of ARGs in the environment

(Wu et al., 2018; Ben Yahia et al., 2018). For instance, the Rooks coming to the Czech Republic in winter were evidenced as reservoirs/vectors of Class-1 integrons (*aadA*), which spreaded AMR to longer distances throughout Europe (Literák et al., 2007). Curiously, a study on wild birds in Tunisia reported that there were Class-1 integrons without GCs (Ben Yahia et al., 2018). The analysis of fecal *E. coli* from wild birds (*Fringila coelebs moreletti*, *Erithacus rubecula*, *Carduelis*, etc) and game birds (common snipe and quail) showed that the Class-1 integrons with multiple GCs conferred resistance to ampicillin, sulfamethoxazole/trimethoprim, tetracyclines, tobramycin, and streptomycin in the Azores (Santos et al., 2013). The owls from the Southern Cone of America were reported to be involved in the spread of AMR (Fuentes-Castillo et al., 2019). On the other hand, the aquatic bird, egret, was found involved in the transmission of ARGs from a contaminated waterway of Jin river into the surrounding Park in Chengdu, although the abundance of ARGs decreased along the assumed transport route (Wu et al., 2018).

The reports of AMR from the wild animals were alarming because of human anthropogenic actions (Radhouani et al., 2014). For instance, from gut microflora of wild and captive fruit bats, Class-1 integrons were conferring resistance to trimethoprim, aminoglycosides, and β-lactams (McDougall et al., 2019). Similarly, Class-1 integrons of *E. coli* isolates from captive giant panda have been identified with ARGs conferring resistance to trimethoprim, streptomycin and spectinomycin (Zhu et al., 2020). Further, the wild boars played a significant role in transmission of AMR in Europe (Torres et al., 2020). Moreover, wild boars, roe deer, red deer, and small mammals such as wood mice, black rats, European rabbits and greater white-toothed shrews, which lived in natural environmental settings with or without lower antibiotics selective pressure, were found to be positive for Class-1 integrons conferring high resistance to fluoroquinolones and tetracyclines (Alonso et al., 2016; Velhner et al., 2018).

8. Reservoirs involved in the spread of resistant *E. coli* to the environment

Development and a continuous persistence of AMR was an alarming threat both for animals and humans (Collignon and McEwen, 2019; Theobald et al., 2019). Multiple reservoirs were involved in the spread of ARGs associated with *E. coli* to the environment (Supplementary Table S1). The plasmid-mediated transfer of ARGs in *E. coli* has been reported from poultry and labor working at the same farm (Dandachi et al., 2019). Many reports have been documented that chicken can become the potential reservoir of resistant bacteria, particularly resistant *E. coli* (Dandachi et al., 2019; Ramadan et al., 2018; Hussain et al., 2017; Stromberg et al., 2017). A survey conducted in the USA on poultry, dairy, and beef farms indicated that both pets and flies living in or around the farms harbored resistant *E. coli* (Poudel et al., 2019). According to another comprehensive survey from 2002 to 2008 conducted by National Antimicrobial Resistance Monitoring System USA, the results revealed the highest level of ARGs in chicken and turkey, followed by beef and pork (Zhao et al., 2012). With a history of resistant *E. coli* in parent flocks, vertical transmission of plasmid-encoded cephamycinases has been reported from day-old hatchery chicks, which corresponded to the same resistant *E. coli* (Projahn et al., 2017). The poultry meat was reported as a reservoir of plasmid-mediated AmpC producing *E. coli* in Tunisia (Maamar et al., 2016). In another study from Nicaragua, the prevalence of resistant *E. coli* was reported with multiple reservoirs such as wild birds, domestic poultry, and farm workers (Hasan et al., 2016). Pigeons have been reported as a reservoir of resistant *E. coli*, and their roles in dissemination of ARGs have also been well explained by conjugation experiments and plasmid analysis (Yang et al., 2015). House flies from a broiler chicken farm have also been reported as a reservoir of ARGs, which contributed to the spread of ARGs to the environment (Solà-Ginés et al., 2015). A cross-sectional study among animals, including broiler, cattle, pigs, and humans, revealed resistant *E. coli* strains were circulating among them (Valentin et al., 2014). With a history of direct contact with chickens at poultry farms, farm workers were reported to be involved in the transmission of resistant bacteria in the Netherlands (Huijbers et al., 2014). In a pattern similar to this, a study from China also recorded MDR among cats, dogs and poultry living nearby (Liao et al., 2013). Food-producing waterfowls (ducks and geese) were shown to be the reservoir of resistant *E. coli* strains, and thus they might be considered as the potential source of ARGs, which can spread resistance genes easily in the farm environment (Zhang et al., 2019). Altogether, the data showed that varieties of reservoirs (flies, dogs, and wild birds) were supposed to be involved in the dissemination of drug-resistant *E. coli* in the environment (Wang et al., 2017).

9. Causes of antibiotics resistance and potential negative implications

It is well known that the production and irregular use of antibiotics have led to rising serious antibiotic resistance problems in the worldwide, although antibiotics have improved the growth performance of livestock as well as health condition of diseased animals and humans. Meanwhile, bacteria are counteracting the effects of potent drugs via evolving mechanisms associated to integrons, plasmids and transposons, which can transfer between the same or different kinds of bacteria, and therefore emerging AMR (Frye and Jackson, 2013). The situation became further worse with the irrational use of antibiotics to cure infections in clinical settings (Qiao et al., 2018). The β-lactams, tetracyclines, streptomycin, erythromycins, sulphonamides, and trimethoprim were a few of the first line of treatment against *E. coli* infections (Holten and Onusko, 2000; Theobald et al., 2019). However, recent studies have reported a varying percentage of AMR against these antibiotics, such as 95.25% for tetracyclines, 91.25% for ampicillin, 84.75% for erythromycin, 88.25% for streptomycin and 65.5% for trimethoprim (Azad et al., 2019; Theobald et al., 2019). Hence, irrational use of such antibiotics without recommended dosage or therapeutic guidelines was the main cause of failing drugs or reducing effectiveness (Asai et al., 2011). For instance, a cross-sectional study was performed in children under five-years old, who were infected with a respiratory disease, diarrhea, and malaria from low- and middle-income countries. It was concluded from ten-years data (2006–2016) that children were treated with unnecessary high number of antibiotics. Thus, to prevent side effects of antibiotics and a risk of emergence of AMR worldwide, a planning is mandatory to reduce antibiotics burden in children (Fink et al., 2020). Without knowing the exact cause of the disease, such a practice will not only increase the cost of cure but also lead to the dissemination of ARGs (Qiao et al., 2018; Theobald et al., 2019). This empirical approach was giving a positive pressure to pathogenic and commensal strains of *E. coli* at the genetic level, leading to the emergence of AMR (Roth et al., 2019).

The trend of using a sub-therapeutic dose of antimicrobials to prevent disease might further facilitate the development of MDR (Yadav and Jha, 2019). The under-dosed antibiotics were usually used in the animal feed, so resident microbiome was under continuous positive antibiotic pressure, resulting in the emergence of drug resistance (Mehdi et al., 2018). The use of antibiotics as a growth promoter is another cause of the development of AMR (Roth et al., 2019; Yadav and Jha, 2019). Most surprisingly, 284 school children aged 8–11 years from Shanghai were investigated about the issues of antibiotic residues. A raised overall detection frequency of many antibiotics was detected in the children, who have higher consumption of poultry meat, livestock, dairy, milk and aquatic products (Wang et al., 2018a). Therefore, it can also be concluded that the misuse of antibiotics is frequent in the field of animal breeding. As such, there is a need to develop novel classes of antimicrobials or to restrict the use of currently available drugs for combating with the worse situation. European Union has already imposed sanctions on the use of antibiotics as a growth promoter in 2006 (Xiong et al., 2018a). Primarily, because of the emergence of novel cassettes among integrons (Piccirillo et al., 2014) and short life span (35 days) in commercial broiler farming, extensive use of antimicrobials as growth promoter was making the condition worse (Mehdi et al., 2018). Nearly, all classes of antimicrobials were used in the agriculture and livestock production system, including penicillins, tetracyclines, aminoglycosides, sulphonamides, quinolones, cephalosporins and colistin (Marshall and Levy, 2011; Catry et al., 2015). The irrational use of different classes of drugs in animals, particularly in food-producing animals, promoted the development of MDR via mobile genetic elements among bacteria. The subsequent transfer of resistant bacteria did occur through the consumption of contaminated food products by humans (Silbergeld et al., 2008; Mehdi et al., 2018), which might cause potential harm to human health. Potential effects and implications of such an under-dosage therapeutics were more in developing countries, particularly in Asia, because of the high population (Mehdi et al., 2018). Meanwhile, there existed a lesser check and balance for the fair use of antibiotics in developing countries, while a lack of research resources and funding further enhanced the bad situation of AMR (Kumar et al., 2019).

As is known to all, antibacterial therapy was a miracle in the 20th century. However, the development of AMR against potent antibiotics is a direct threat to public health (Amann et al., 2019). A rise of AMR was speculated to increase mortality, morbidity, and loss of resources across the globe. There were several studies available, revealing a significant increase in AMR from the last 50 years (Shlaes et al., 1997; Soufi et al., 2011; Piccirillo et al., 2014). The United States Centers for Disease Control and Prevention (CDC) declared that, because of AMR, about 2 million resistant infections and 23,000 deaths per year might occur in the USA, along with the cost of $20 billion in disease management and $35 billion in terms of production losses. Approximately 25,000 deaths per year along with an estimated loss of €1.5 billion annually were expected in Europe (Gelband et al., 2015). Since there is limited research and development in underdeveloped countries, it is difficult to ensure the exact data or estimate direct/indirect economic losses due to AMR. However, an annual death of approximately 58,000 children was predicted in India as a result of AMR (Laxminarayan et al., 2013). The increasing death rate in children was reported from Tanzania and Mozambique, mainly on account of AMR (Fink et al., 2020; Kayange et al., 2010). In these circumstances, *E. coli* is one of the most prevalent bacterium showing resistance to a wide variety of antibiotics, including the fourth-generation cephalosporin. The emerging co-resistance to carbapenems and colistin has been recorded from China, raising immense apprehension that we are at the edge of post antibiotic time (Yao et al., 2016; Zheng et al., 2016). Recently, a clinically important *E. coli* strain from Vietnam carried four different plasmids, conferring resistance to carbapenems and colistin, collectively. Further, the resistant genes of this *E. coli* strain were found to be carried by the complex Class-1 integron containing two complete structures of IS*CR1*, which showed this integron could transfer or spread easily among bacteria (Le-Vo et al., 2019). Therefore, it can be predicted that the therapeutic management of resistant *E. coli* infections will become more difficult. In 2013, a survey data showed that 85%–100% of *E. coli* isolates from 17 European countries were resistant strains (Gelband et al., 2015). The entire situation tells the negative implications of antibiotic resistance and how we are unsafe when facing with these resistant bacteria. Human beings may be at risk, if no measures are taken to control the rising status of AMR (Fig. 1). Bacterial pathogens with various ARGs constitute a significant threat to public health, where the presence of resistant bacteria in food products is directly affecting the health condition of humans. In addition, there was a considerable concern of AMR between many countries and international agencies, so a measure named “One Health” was under active planning, in order to solve the problem of antibiotic resistance (Collignon and McEwen, 2019). In Sep 2017, scientists from 14 countries have identified four critical gaps, which were required to handle and mitigate the harms of antibiotics. The identification of gaps were required for future research and development, to handle AMR. The first gap identified was the relative significance of various sources of antibiotics and drug-resistant bacteria to the environment; secondly, the anthropogenic factors that were involved in the evolution of AMR; thirdly, the effects of resistant bacteria on animals and human health; and fourthly, the success and viability of different social, behavioral, technological and economic intervention, which were essential to diminish environmental AMR (Larsson et al., 2018).

10. Prevention, management, and control of AMR

The use of antimicrobials increased as time went on in different geographical areas, in terms of prophylaxis, metaphylaxis, therapeutics and growth promoter. Approximately two-thirds of total antimicrobial drugs were used in the livestock sector (Gelband et al., 2015). This massive use of antibiotics resulted in the emergence of AMR against the potential anti-microbial drugs (Brower et al., 2017). The integrons were often carried by plasmid, the root cause of AMR among members of *Enterobacteriaceae* in animals, humans, and the environment (Ansari et al., 2008; Sunde et al., 2015). Nationally, necessary measures are ongoing to combat with the AMR. For instance, gentamicin and ceftiofur were previously used for the treatment of gram-negative bacteria. However, from 2003 in Canada, partial restrictions were imposed, resulting in a reduced drug resistance. The results were so convincing that a full ban on the usage of gentamicin and ceftiofur was implemented in poultry by the Canadian government (Chalmers et al., 2017). Similar practices were applied in France, where requirements such as prudent exercise of antibiotics were imposed on hospitals via antibiotic stewardship program (Allerberger et al., 2009). Although the significant reduction in use of antibiotics have been observed in humans (Sabuncu et al., 2009; Huttner and Harbarth, 2009) and veterinary settings (Casella et al., 2017). These efforts were still not so promising, because a large proportion of *E. coli* isolated (91.7%) from retailed chick shops have been confirmed to carry ARGs (Casella et al., 2017). Further, nationwide surveys among acute care hospitals of France indicated antibiotic stewardship programs were not fully implemented. The reason of collapse could be a considerable gap between regulations and its subsequent implementation, or may be due to inadequate institutional support (Binda et al., 2019), and thus a regular monitoring and evaluation of program was recommended (Delannoy et al., 2019).

In addition to Brazil, India, Russia, and South Africa, China is one of the main contributors to the utilization of antimicrobial drugs. It is a commonplace fact that ampicillin, chlortetracycline, oxytetracycline, enrofloxacin, ciprofloxacin, and doxycycline are commonly used drugs, worldwide. Further, it was estimated that the global average use of antibiotics in food-producing animals would increase by 200% by the end of 2030 (Klein et al., 2018). In India, a 312% increase in antimicrobial drug usage was estimated (Van Boeckel et al., 2015). Therefore, some countries were planning to limit the use of antibiotics. For instance, owing to excessive usage of fluoroquinolones between animals and humans as well as its subsequent implication in the development of AMR, the USA have imposed a strict ban on the usage of fluoroquinolones in 2005 (Nelson et al., 2007). The Ministry of Agriculture of China imposed a ban on the use of colistin sulfate as a feed additive in 2016 (Walsh and Wu, 2016) and the Chinese government officially implemented this policy on April 30, 2017 (Wang et al., 2020), and more antibiotics will be expected to be banned by 2020. Further, the India has promised to ban the use of colistin as growth promoter in the livestock sector, and improve infection control management in hospital settings as well as impose antimicrobial stewardship (Laxminarayan et al., 2020).

Encouragingly, there are novel methods or alternates to overcome the problem of MDR, which include specific bacteriophages, lysine, antimicrobial peptides, bacteriocins, and antibodies (Ghosh et al., 2019). As result of the administration of probiotic lyophilized intestinal micro-flora, the significant reduction in the transmission as well as excretion of resistant *E. coli* was observed (Ceccarelli et al., 2017). The study indicated that not only such products can be a source of growth promoter but it can also overcome the problem of AMR. In this case, it is necessary to look for alternative products of antibiotics, such as specific bacteriophages and antimicrobial peptides. Indeed, an application of bacteriophage is safer because it is specific to the particular bacteria and capability to infect specific strain or species. Further, it is not harmful to the commensal bacterial flora (Wernicki et al., 2017). Adding a small amount of clay in the livestock and poultry manure has been evidenced to reduce the relative abundances of ARGs (Awasthi et al., 2019). Converting the pig manure into biochar would reduce the abundance of ARGs in soil, compared with using it as organic fertilizer directly (Zhou et al., 2019b).

For proper control and prevention of AMR issue, the followings need to be noticed (1) spread and prevention of disease (Furusawa et al., 2018); (2) intervention strategies for resistance pattern, evaluation through continuous reporting (Tacconelli et al., 2018b); and (3) appropriate legislations and restrictions on the use of antibiotics, especially in food-producing animals (Bertollo et al., 2018; Wang et al., 2020). Further, it is required to develop novel antibiotics or improve the efficacy of available antibiotics (Tacconelli et al., 2018a). In addition, limited use of the novel antibiotics should be considered to avoid emergence and dissemination of upcoming resistance to these drugs (Denny et al., 2020). Moreover, the advances in diagnostic tools for precise and accurate diagnosis (Furusawa et al., 2018), precision livestock farming, as well as proper management of animal houses are required to prevent infection and improve animal health (Tullo et al., 2019). Strict monitoring of water resources is also necessary, as ARGs can spread through water environment (Zhou et al., 2018). The practicing veterinarian should be well informed and updated about the prevailing ARGs in the vicinity (Speksnijder and Wagenaar, 2018).

Cumulating together, in order to control and restrict the use of antimicrobials, there is an urgent need to design rules and regulations by countries and international agencies. This is far important, especially for underdeveloped countries where contracted and seasonal farmers exist in the market. These farmers were taking animal shed/farms on rent and followed practices of excessive usage of antimicrobials instead of focusing on bio-security and relevant disinfection procedures. Hence, it is essential to communicate with farmers and workers for bio-security related measures at the farm settings. In addition, strict check and balance should be imposed to veterinary practitioners for the use of antibiotics (Speksnijder and Wagenaar, 2018). To handle and restrict the transmission of AMR in the environment, the mechanisms involved in the transfer of ARGs should be investigated, firstly. Moreover, immediate steps should be taken to stop the spread of AMR, which included the restricted use of antibiotics in clinical settings and removal of ARGs from drinking water and agricultural system (Zhou et al., 2018; Qiao et al., 2018).

11. Conclusion and prospects

No matter the dosage is insufficient or prescription is wrong, a positive pressure caused by the irrational use of antibiotics can promote the development and dissemination of ARGs among the environment, including WWTP in or around livestock farms. In this regard, the integrons or complex integrons are believed to be the basic unit and one of important cause for the transfer of genetic resistance. A variety of reservoirs are involved in the environment by spreading of ARGs, which includes livestock, flies, pigs, farm workers, waterfowls, wild birds, poultry meat, free-range poultry, farm waste, WWTP, and agricultural land. Alongside environment-control legislative measures, a systematic evidence-based diagnosis or prescription should be maintained by human and veterinary clinicians with a proper digital record.

Furthermore, while focusing on nexus of standardization and outcome, a continuous monitoring for ARGs among various reservoirs from different parts of the globe is urgent and essential. A central networking is required for proper monitoring of records and its maintenance. A more appropriate research funding, system of report for writing matrix, analytical methodologies, and the necessary infrastructure should be established to control the harmfulness of AMR/MDR worldwide. In a word, continuous research funding, surveillance and prevalence reporting mechanism are required for proper understanding of prevalent classes of integrons in *E. coli* strains. This will contribute to the development of preventive measures and novel effective antimicrobial drugs according to the needs of the times.

CRediT authorship contribution statement

Shaqiu Zhang: Conceptualization, Writing - original draft, Writing - review & editing, Funding acquisition. **Muhammad Abbas:** Writing - original draft, Writing - review & editing, Visualization. **Mujeeb Ur Rehman:** Writing - review & editing, Visualization. **Yahui Huang:** Writing - review & editing. **Rui Zhou:** Data curation. **Siyue Gong:** Data curation. **Hong Yang:** Writing - review & editing. **Shuling Chen:** Writing - review & editing. **Mingshu Wang:** Funding acquisition. **Anchun Cheng:** Writing - review & editing, Funding acquisition, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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References

Adelowo, O.O., Fagade, O.E., Agersø, Y., 2014. Antibiotic resistance and resistance genes in *Escherichia coli* from poultry farms, southwest Nigeria. J. Infect. Dev. Ctries. 8, 1103–1112. https://doi.org/10.3855/jidc.4222.

Alam, M.U., Rahman, M., Abdullah-Al-Masud, Islam, M.A., Asaduzzaman, M., Sarker, S., Rousham, E., Unicomb, L., 2019. Human exposure to antimicrobial resistance from poultry production: assessing hygiene and waste-disposal practices in Bangladesh. Int. J. Hyg Environ. Health 222, 1068–1076. https://doi.org/10.1016/j.ijheh.2019.07.007.

Ali, T., ur Rahman, S., Zhang, L., Shahid, M., Zhang, S., Liu, G., Gao, J., Han, B., 2016. ESBL-producing *Escherichia coli* from cows suffering mastitis in China contain clinical class 1 integrons with CTX-M linked to IS*CR1*. Front. Microbiol. 7 https://doi.org/10.3389/fmicb.2016.01931.

Allerberger, F., Gareis, R., Jindrák, V., Struelens, M.J., 2009. Antibiotic stewardship implementation in the EU: the way forward. Expert Rev. Anti Infect. Ther. 7, 1175–1183. https://doi.org/10.1586/eri.09.96.

Alonso, C.A., González-Barrio, D., Tenorio, C., Ruiz-Fons, F., Torres, C., 2016. Antimicrobial resistance in faecal *Escherichia coli* isolates from farmed red deer and wild small mammals. Detection of a multiresistant *E. coli* producing extended-spectrum beta-lactamase. Comp. Immunol. Microbiol. Infect. Dis. 45, 34–39. https://doi.org/10.1016/j.cimid.2016.02.003.

Alonso, C.A., Cortés-Cortés, G., Maamar, E., Massó, M., Rocha-Gracia, R., del, C., Torres, C., Centrón, D., Quiroga, M.P., 2018. Molecular diversity and conjugal transferability of class 2 integrons among *Escherichia coli* isolates from food, animal and human sources. Int. J. Antimicrob. Agents 51, 905–911. https://doi.org/10.1016/j.ijantimicag.2018.02.001.

Amador, P., Fernandes, R., Prudêncio, C., Duarte, I., 2019. Prevalence of antibiotic resistance genes in multidrug-resistant *Enterobacteriaceae* on Portuguese livestock manure. Antibiotics 8, 23. https://doi.org/10.3390/antibiotics8010023.

Amann, S., Neef, K., Kohl, S., 2019. Antimicrobial resistance (AMR). Eur. J. Hosp. Pharm. 26, 175–177. https://doi.org/10.1136/ejhpharm-2018-001820.

Ansari, M.I., Grohmann, E., Malik, A., 2008. Conjugative plasmids in multi-resistant bacterial isolates from Indian soil. J. Appl. Microbiol. 104, 1774–1781. https://doi.org/10.1111/j.1365-2672.2008.03736.x.

Asai, T., Masani, K., Sato, C., Hiki, M., Usui, M., Baba, K., Ozawa, M., Harada, K., Aoki, H., Sawada, T., 2011. Phylogenetic groups and cephalosporin resistance genes of *Escherichia coli* from diseased food-producing animals in Japan. Acta Vet. Scand. 53, 52. https://doi.org/10.1186/1751-0147-53-52.

Awad, A., Arafat, N., Elhadidy, M., 2016. Genetic elements associated with antimicrobial resistance among avian pathogenic *Escherichia coli*. Ann. Clin. Microbiol. Antimicrob. 15, 59. https://doi.org/10.1186/s12941-016-0174-9.

Awasthi, M.K., Liu, T., Chen, H., Verma, S., Duan, Y., Awasthi, S.K., Wang, Q., Ren, X., Zhao, J., Zhang, Z., 2019. The behavior of antibiotic resistance genes and their associations with bacterial community during poultry manure composting. Bioresour. Technol. 280, 70–78. https://doi.org/10.1016/j.biortech.2019.02.030.

Azad, M.A.R.A., Rahman, M.M., Amin, R., Begum, M.I.A., Fries, R., Husna, A., Khairalla, A.S., Badruzzaman, A.T.M., El Zowalaty, M.E., Lampang, K.N., Ashour, H.M., Hafez, H.M., 2019. Susceptibility and multidrug resistance patterns of *Escherichia coli* isolated from cloacal swabs of live broiler chickens in Bangladesh. Pathog. Basel Switz. 8, 118. https://doi.org/10.3390/pathogens8030118.

Badura, A., Luxner, J., Feierl, G., Reinthaler, F.F., Zarfel, G., Galler, H., Pregartner, G., Riedl, R., Grisold, A.J., 2014. Prevalence, antibiotic resistance patterns and molecular characterization of *Escherichia coli* from Austrian sandpits. Environ. Pollut. 194, 24–30. https://doi.org/10.1016/j.envpol.2014.07.007.

Bae, I.K., Lee, Y.H., Jeong, H.J., Hong, S.G., Lee, S.H., Jeong, S.H., 2008. A novel $bla_{CTX-M-14}$ gene-harboring complex class 1 integron with an In4-like backbone structure from a clinical isolate of *Escherichia coli*. Diagn. Microbiol. Infect. Dis. 62, 340–342. https://doi.org/10.1016/j.diagmicrobio.2008.06.006.

Baron, S.A., Diene, S.M., Rolain, J.M., 2018. Human microbiomes and antibiotic resistance. Hum. Microbiome J. 10, 43–52. https://doi.org/10.1016/j.humic.2018.08.005.

Bélanger, L., Garenaux, A., Harel, J., Boulianne, M., Nadeau, E., Dozois, C.M., 2011. *Escherichia coli* from animal reservoirs as a potential source of human extra-intestinal pathogenic *E. coli*. FEMS Immunol. Med. Microbiol. 62, 1–10. https://doi.org/10.1111/j.1574-695X.2011.00797.x.

Ben Yahia, H., Ben Sallem, R., Tayh, G., Klibi, N., Ben Amor, I., Gharsa, H., Boudabbous, A., Ben Slama, K., 2018. Detection of CTX-M-15 harboring *Escherichia coli* isolated from wild birds in Tunisia. BMC Microbiol. 18, 26. https://doi.org/10.1186/s12866-018-1163-2.

Bertollo, L.G., Lutkemeyer, D.S., Levin, A.S., 2018. Are antimicrobial stewardship programs effective strategies for preventing antibiotic resistance? A systematic review. Am. J. Infect. Contr. 46, 824–836. https://doi.org/10.1016/j.ajic.2018.01.002.

Binda, F., Tebano, G., Kallen, M.C., ten Oever, J., Hulscher, M.E., Schouten, J.A., Pulcini, C., 2019. Nationwide survey of hospital antibiotic stewardship programs in France. Med. Maladies Infect. https://doi.org/10.1016/j.medmal.2019.09.007.

Brower, C.H., Mandal, S., Hayer, S., Sran, M., Zehra, A., Patel, S.J., Kaur, R., Chatterjee, L., Mishra, S., Das, B.R., Singh, P., Singh, R., Gill, J.P.S., Laxminarayan, R., 2017. The prevalence of extended-spectrum beta-lactamase-producing multidrug-resistant *Escherichia coli* in poultry chickens and variation according to farming practices in Punjab, India. Environ. Health Perspect. 125 https://doi.org/10.1289/EHP292, 077015-1-077015-10.

Calero-Cáceres, W., Méndez, J., Martín-Díaz, J., Muniesa, M., 2017. The occurrence of antibiotic resistance genes in a Mediterranean river and their persistence in the riverbed sediment. Environ. Pollut. 223, 384–394. https://doi.org/10.1016/j.envpol.2017.01.035.

Cameron, F.H., Groot Obbink, D.J., Ackerman, V.P., Hall, R.M., 1986. Nucleotide sequence of the AAD (2'') aminoglycoside adenylyltransferase determinant *aadB*. Evolutionary relationship of this region with those surrounding *aadA* in R538-1 and *dhfrII* in R388. Nucleic Acids Res. 14, 8625–8635.

Canal, N., Meneghetti, K.L., de Almeida, C.P., da Rosa Bastos, M., Otton, L.M., Corção, G., 2016. Characterization of the variable region in the class 1 integron of antimicrobial-resistant *Escherichia coli* isolated from surface water. Braz. J. Microbiol. 47, 337–344. https://doi.org/10.1016/j.bjm.2016.01.015.

Casella, T., Rodríguez, M.M., Takahashi, J.T., Ghiglione, B., Dropa, M., Assunção, E., Nogueira, M.L., Lincopan, N., Gutkind, G., Nogueira, M.C.L., 2015. Detection of bla_{CTX-M}-type genes in complex class 1 integrons carried by *Enterobacteriaceae* isolated from retail chicken meat in Brazil. Int. J. Food Microbiol. 197, 88–91. https://doi.org/10.1016/j.ijfoodmicro.2014.12.001.

Casella, T., Nogueira, M.C.L., Saras, E., Haenni, M., Madec, J.-Y., 2017. High prevalence of ESBLs in retail chicken meat despite reduced use of antimicrobials in chicken production, France. Int. J. Food Microbiol. 257, 271–275. https://doi.org/10.1016/j.ijfoodmicro.2017.07.005.

Catry, B., Cavaleri, M., Baptiste, K., Grave, K., Grein, K., Holm, A., Jukes, H., Liebana, E., Lopez Navas, A., Mackay, D., Magiorakos, A.-P., Moreno Romo, M.A., Moulin, G., Muñoz Madero, C., Matias Ferreira Pomba, M.C., Powell, M., Pyörälä, S., Rantala, M., Ružauskas, M., Sanders, P., Teale, C., Threlfall, E.J., Törneke, K., van Duijkeren, E., Torren Edo, J., 2015. Use of colistin-containing products within the European Union and European Economic Area (EU/EEA): development of resistance in animals and possible impact on human and animal health. Int. J. Antimicrob. Agents 46, 297–306. https://doi.org/10.1016/j.ijantimicag.2015.06.005.

Ceccarelli, D., van Essen-Zandbergen, A., Smid, B., Veldman, K.T., Boender, G.J., Fischer, E.A.J., Mevius, D.J., van der Goot, J.A., 2017. Competitive exclusion reduces transmission and excretion of extended-spectrum-β-lactamase-producing *Escherichia coli* in broilers. Appl. Environ. Microbiol. 83 (16), e03439 https://doi.org/10.1128/AEM.03439-16.

Chalmers, G., Cormier, A.C., Nadeau, M., Côté, G., Reid-Smith, R.J., Boerlin, P., 2017. Determinants of virulence and of resistance to ceftiofur, gentamicin, and spectinomycin in clinical *Escherichia coli* from broiler chickens in Québec, Canada. Vet. Microbiol. 203, 149–157. https://doi.org/10.1016/j.vetmic.2017.02.005.

Chandler, J.C., Aljasir, S.F., Hamidi, A., Sylejmani, D., Gerow, K.G., Bisha, B., 2018. Short communication: a countrywide survey of antimicrobial-resistant indicator bacteria in Kosovo's dairy farms. J. Dairy Sci. 101, 6982–6989. https://doi.org/10.3168/jds.2017-14091.

Chen, Q., An, X., Li, H., Su, J., Ma, Y., Zhu, Y.G., 2016. Long-term field application of sewage sludge increases the abundance of antibiotic resistance genes in soil. Environ. Int. 92– 93, 1–10. https://doi.org/10.1016/j.envint.2016.03.026.

Chen, C., Pankow, C.A., Oh, M., Heath, L.S., Zhang, L., Du, P., Xia, K., Pruden, A., 2019. Effect of antibiotic use and composting on antibiotic resistance gene abundance and resistome risks of soils receiving manure-derived amendments. Environ. Int. 128, 233–243. https://doi.org/10.1016/j.envint.2019.04.043.

Cheng, W., Li, J., Wu, Y., Xu, L., Su, C., Qian, Y., Zhu, Y.G., Chen, H., 2016. Behavior of antibiotics and antibiotic resistance genes in eco-agricultural system: a case study. J. Hazard Mater. 304, 18–25. https://doi.org/10.1016/j.jhazmat.2015.10.037.

Chern, E.C., Brenner, K.P., Wymer, L., Haugland, R.A., 2009. Comparison of fecal indicator bacteria densities in marine recreational waters by QPCR. Water Qual. Expo. Health 1, 203–214. https://doi.org/10.1007/s12403-009-0019-2.

Cho, S., Nguyen, H.A.T., McDonald, J.M., Woodley, T.A., Hiott, L.M., Barrett, J.B., Jackson, C.R., Frye, J.G., 2019. Genetic characterization of antimicrobial-resistant *Escherichia coli* isolated from a mixed-use watershed in northeast Georgia, USA. Int. J. Environ. Res. Publ. Health 16 (19), 3761. https://doi.org/10.3390/ijerph16193761.

Collignon, P.J., McEwen, S.A., 2019. One health—its importance in helping to better control antimicrobial resistance. Trav. Med. Infect. Dis. 4, 22. https://doi.org/10.3390/tropicalmed4010022.

Collis, C.M., Hall, R.M., 1995. Expression of antibiotic resistance genes in the integrated cassettes of integrons. Antimicrob. Agents Chemother. 39, 155–162.

Collis, C.M., Kim, M.J., Partridge, S.R., Stokes, H.W., Hall, R.M., 2002. Characterization of the class 3 integron and the site-specific recombination system it determines. J. Bacteriol. 184, 3017–3026. https://doi.org/10.1128/JB.184.11.3017-3026.2002.

Dandachi, I., Fayad, E., Sleiman, A., Daoud, Z., Rolain, J.M., 2019. Dissemination of multidrug-resistant and *mcr-1* gram-negative bacilli in broilers, farm workers, and the surrounding environment in Lebanon. Microb. Drug Resist. 26, 368–377. https://doi.org/10.1089/mdr.2019.0137.

Davies, J., 1994. Inactivation of antibiotics and the dissemination of resistance genes. Science 264, 375–382. https://doi.org/10.1126/science.8153624.
Delannoy, M., Agrinier, N., Charmillon, A., Degand, N., Dellamonica, J., Leone, M., Pulcini, C., Novy, E., 2019. Implementation of antibiotic stewardship programmes in French ICUs in 2018: a nationwide cross-sectional survey. J. Antimicrob. Chemother. 74, 2106–2114. https://doi.org/10.1093/jac/dkz113.
Deng, Y., Bao, X., Ji, L., Chen, L., Liu, J., Miao, J., Chen, D., Bian, H., Li, Y., Yu, G., 2015. Resistance integrons: class 1, 2 and 3 integrons. Ann. Clin. Microbiol. Antimicrob. 14, 45. https://doi.org/10.1186/s12941-015-0100-6.
Denny, K.J., De Wale, J., Laupland, K.B., Harris, P.N.A., Lipman, J., 2020. When not to start antibiotics: avoiding antibiotic overuse in the intensive care unit. Clin. Microbiol. Infect. 26, 35–40. https://doi.org/10.1016/j.cmi.2019.07.007.
Dessie, H.K., Bae, D.H., Lee, Y.J., 2013. Characterization of integrons and their cassettes in *Escherichia coli* and *Salmonella* isolates from poultry in Korea. Poultry Sci. 92, 3036–3043. https://doi.org/10.3382/ps.2013-03312.
Devarajan, N., Laffite, A., Graham, N.D., Meijer, M., Prabakar, K., Mubedi, J.I., Elongo, V., Mpiana, P.T., Ibelings, B.W., Wildi, W., Poté, J., 2015. Accumulation of clinically relevant antibiotic-resistance genes, bacterial load, and metals in freshwater lake sediments in central Europe. Environ. Sci. Technol. 49, 6528–6537. https://doi.org/10.1021/acs.est.5b01031.
Devarajan, N., Laffite, A., Mulaji, C.K., Otamonga, J.P., Mpiana, P.T., Mubedi, J.I., Prabakar, K., Ibelings, B.W., Poté, J., 2016. Occurrence of antibiotic resistance genes and bacterial markers in a tropical river receiving hospital and urban wastewaters. PloS One 11. https://doi.org/10.1371/journal.pone.0149211.
Dhawde, R., Macaden, R., Saranath, D., Nilgiriwala, K., Ghadge, A., Birdi, T., 2018. Antibiotic resistance characterization of environmental *E. coli* isolated from river mula-mutha, pune district, India. Int. J. Environ. Res. Publ. Health 15, 1247. https://doi.org/10.3390/ijerph15061247.
Dho-Moulin, M., Fairbrother, J.M., 1999. Avian pathogenic *Escherichia coli* (APEC). Vet. Res. 30, 299–316.
Dou, X., Gong, J., Han, X., Xu, M., Shen, H., Zhang, D., Zhuang, L., Liu, J., Zou, J., 2016. Characterization of avian pathogenic *Escherichia coli* isolated in eastern China. Gene 576, 244–248. https://doi.org/10.1016/j.gene.2015.10.012.
Duan, M., Gu, J., Wang, X., Li, Y., Zhang, R., Hu, T., Zhou, B., 2019. Factors that affect the occurrence and distribution of antibiotic resistance genes in soils from livestock and poultry farms. Ecotoxicol. Environ. Saf. 180, 114–122. https://doi.org/10.1016/j.ecoenv.2019.05.005.
Economou, V., Gousia, P., 2015. Agriculture and food animals as a source of antimicrobial-resistant bacteria. Infect. Drug Resist. 8, 49–61. https://doi.org/10.2147/IDR.S55778.
El-Demerdash, A.S., Aggour, M.G., El-Azzouny, M.M., Abou-Khadra, S.H., 2018. Molecular analysis of integron gene cassette arrays associated multi-drug resistant *Enterobacteriaceae* isolates from poultry. Cell. Mol. Biol. Noisy–Gd. Fr. 64, 149–156.
Elizabeth, R., Chanda, D.D., Chakravarty, A., Paul, D., Chetri, S., Bhowmik, D., Wangkheimayum, J., Bhattacharjee, A., 2018. Association of glycerol kinase gene with class 3 integrons: a novel cassette array within *Escherichia coli*. Indian J. Med. Microbiol. 36, 104. https://doi.org/10.4103/ijmm.IJMM_17_188.
Escudero, J.A., Loot, C., Nivina, A., Mazel, D., 2015. The integron: adaptation on demand. Microbiol. Spectr. 3, 2. https://doi.org/10.1128/microbiolspec.MDNA3-0019-2014.
Fakhr, A.E., Gohar, M.K., Atta, A.H., 2016. Impact of some ecological factors on fecal contamination of drinking water by diarrheagenic antibiotic-resistant *Escherichia coli* in zagazig city, Egypt. Internet J. Microbiol. 2016 https://doi.org/10.1155/2016/6240703.
Fink, G., D'Acremont, V., Leslie, H.H., Cohen, J., 2020. Antibiotic exposure among children younger than 5 years in low-income and middle-income countries: a cross-sectional study of nationally representative facility-based and household-based surveys. Lancet Infect. Dis. 20, 179–187. https://doi.org/10.1016/S1473-3099(19)30572-9.
Frye, J.G., Jackson, C.R., 2013. Genetic mechanisms of antimicrobial resistance identified in *Salmonella enterica, Escherichia coli*, and *Enteroccocus* spp. isolated from U.S. food animals. Front. Microbiol. 4, 135. https://doi.org/10.3389/fmicb.2013.00135.
Fuentes-Castillo, D., Farfán-López, M., Esposito, F., Moura, Q., Fernandes, M.R., Lopes, R., Cardoso, B., Muñoz, M.E., Cerdeira, L., Najle, I., Muñoz, P.M., Catão-Dias, J.L., González-Acuña, D., Lincopan, N., 2019. Wild owls colonized by international clones of extended-spectrum β-lactamase (CTX-M)-producing *Escherichia coli* and *Salmonella Infantis* in the Southern Cone of America. Sci. Total Environ. 674, 554–562. https://doi.org/10.1016/j.scitotenv.2019.04.149.
Furusawa, C., Horinouchi, T., Maeda, T., 2018. Toward prediction and control of antibiotic-resistance evolution. Curr. Opin. Biotechnol., Analytical Biotechnology 54, 45–49. https://doi.org/10.1016/j.copbio.2018.01.026.
Garner, E., Benitez, R., von Wagoner, E., Sawyer, R., Schaberg, E., Hession, W.C., Krometis, L.-A.H., Badgley, B.D., Pruden, A., 2017. Stormwater loadings of antibiotic resistance genes in an urban stream. Water Res. 123, 144–152. https://doi.org/10.1016/j.watres.2017.06.046.
Gelband, H., Molly Miller, P., Pant, S., Gandra, S., Levinson, J., Barter, D., White, A., Laxminarayan, R., 2015. The state of the world's antibiotics 2015. Wound Heal. South. Afr. 8, 30–34.
Ghosh, C., Sarkar, P., Issa, R., Haldar, J., 2019. Alternatives to conventional antibiotics in the era of antimicrobial resistance. Trends Microbiol. 27, 323–338. https://doi.org/10.1016/j.tim.2018.12.010.
Gillings, M.R., 2013. Evolutionary consequences of antibiotic use for the resistome, mobilome and microbial pangenome. Front. Microbiol. 4, 4. https://doi.org/10.3389/fmicb.2013.00004.
Gillings, M.R., 2014. Integrons: past, present, and future. Microbiol. Mol. Biol. Rev. 78, 257–277. https://doi.org/10.1128/MMBR.00056-13.
Gillings, M.R., Holley, M.P., Stokes, H.W., Holmes, A.J., 2005. Integrons in Xanthomonas: a source of species genome diversity. Proc. Natl. Acad. Sci. Unit. States Am. 102, 4419–4424. https://doi.org/10.1073/pnas.0406620102.
Gillings, M., Boucher, Y., Labbate, M., Holmes, A., Krishnan, S., Holley, M., Stokes, H.W., 2008. The evolution of class 1 integrons and the rise of antibiotic resistance. J. Bacteriol. 190, 5095–5100. https://doi.org/10.1128/JB.00152-08.
Guo, Q., Ding, B., Jové, T., Stoesser, N., Cooper, V.S., Wang, M., Doi, Y., 2016. Characterization of a novel IncHI2 plasmid carrying tandem copies of $bla_{CTX-M-2}$ in a *fosA6*-harboring *Escherichia coli* ST410 strain. Antimicrob. Agents Chemother. 60, 6742–6747. https://doi.org/10.1128/AAC.01173-16.
Han, X., Hu, H., Chen, Q., Yang, L., Li, H., Zhu, Y., Li, X., Ma, Y., 2018. Antibiotic resistance genes and associated bacterial communities in agricultural soils amended with different sources of animal manures. Soil Biol. Biochem. 126, 91–102. https://doi.org/10.1016/j.soilbio.2018.08.018.
Harada, K., Asai, T., 2010. Role of antimicrobial selective pressure and secondary factors on antimicrobial resistance prevalence in *Escherichia coli* from food-producing animals in Japan. J. Biomed. Biotechnol. 2010 https://doi.org/10.1155/2010/180682.
Harakeh, S., Yassine, H., El-Fadel, M., 2006. Antimicrobial-resistant patterns of *Escherichia coli* and *Salmonella* strains in the aquatic Lebanese environments. Environ. Pollut. 143, 269–277. https://doi.org/10.1016/j.envpol.2005.11.027.
Harmse, J.L., Engelbrecht, J.C., Bekker, J.L., 2016. The impact of physical and ergonomic hazards on poultry abattoir processing workers: a review. Int. J. Environ. Res. Publ. Health 13, 197. https://doi.org/10.3390/ijerph13020197.
Hasan, B., Laurell, K., Rakib, M.M., Ahlstedt, E., Hernandez, J., Caceres, M., Järhult, J.D., 2016. Fecal carriage of extended-spectrum β-lactamases in healthy humans, poultry, and wild birds in león, Nicaragua-A shared pool of bla_{CTX-M} genes and possible interspecies clonal spread of extended-spectrum β-lactamases-producing *Escherichia coli*. Microb. Drug Res. 22, 682–687. https://doi.org/10.1089/mdr.2015.0323.
He, L., Ying, G., Liu, Y., Su, H., Chen, J., Liu, S., Zhao, J., 2016. Discharge of swine wastes risks water quality and food safety: antibiotics and antibiotic resistance genes from swine sources to the receiving environments. Environ. Int. 92–93, 210–219. https://doi.org/10.1016/j.envint.2016.03.023.
He, T., Wang, R., Liu, D., Walsh, T.R., Zhang, R., Lv, Y., Ke, Y., Ji, Q., Wei, R., Liu, Z., Shen, Y., Wang, G., Sun, L., Lei, L., Lv, Z., Li, Y., Pang, M., Wang, L., Sun, Q., Fu, Y., Song, H., Hao, Y., Shen, Z., Wang, S., Chen, G., Wu, C., Shen, J., Wang, Y., 2019. Emergence of plasmid-mediated high-level tigecycline resistance genes in animals and humans. Nat. Microbiol. 4, 1450–1456. https://doi.org/10.1038/s41564-019-0445-2.
Headey, D., Hirvonen, K., 2016. Is exposure to poultry harmful to child nutrition? An observational analysis for rural Ethiopia. PloS One 11. https://doi.org/10.1371/journal.pone.0160590.
Henchion, M., McCarthy, M., Resconi, V.C., Troy, D., 2014. Meat consumption: trends and quality matters. Meat Sci. 98, 561–568. https://doi.org/10.1016/j.meatsci.2014.06.007.
Heuer, H., Focks, A., Lamshöft, M., Smalla, K., Matthies, M., Spiteller, M., 2008. Fate of sulfadiazine administered to pigs and its quantitative effect on the dynamics of bacterial resistance genes in manure and manured soil. Soil Biol. Biochem. 40, 1892–1900. https://doi.org/10.1016/j.soilbio.2008.03.014.
Heuer, H., Schmitt, H., Smalla, K., 2011. Antibiotic resistance gene spread due to manure application on agricultural fields. Curr. Opin. Microbiol. 14, 236–243. https://doi.org/10.1016/j.mib.2011.04.009.
Heuer, H., Binh, C.T.T., Jechalke, S., Kopmann, C., Zimmerling, U., Krögerrecklenfort, E., Ledger, T., Gonzalez, B., Top, E.M., Smalla, K., 2012. IncP-1ε plasmids are important vectors of antibiotic resistance genes in agricultural systems: diversification driven by class 1 integron gene cassettes. Front. Microbiol. 3, 2. https://doi.org/10.3389/fmicb.2012.00002.
Holten, K.B., Onusko, E.M., 2000. Appropriate prescribing of oral beta-lactam antibiotics. Am. Fam. Physician 62, 611–620.
Horner, C., Mawer, D., Wilcox, M., 2012. Reduced susceptibility to chlorhexidine in staphylococci: is it increasing and does it matter? J. Antimicrob. Chemother. 67, 2547–2559. https://doi.org/10.1093/jac/dks284.
Hsu, J.T., Chen, C.Y., Young, C.W., Chao, W.L., Li, M.H., Liu, Y.H., Lin, C.M., Ying, C., 2014. Prevalence of sulfonamide-resistant bacteria, resistance genes and integron-associated horizontal gene transfer in natural water bodies and soils adjacent to a swine feedlot in northern Taiwan. J. Hazard Mater. 277, 34–43. https://doi.org/10.1016/j.jhazmat.2014.02.016.
Huijbers, P.M.C., Graat, E.a.M., Haenen, A.P.J., van Santen, M.G., van Essen-Zandbergen, A., Mevius, D.J., van Duijkeren, E., van Hoek, A.H.a.M., 2014. Extended-spectrum and AmpC β-lactamase-producing *Escherichia coli* in broilers and people living and/or working on broiler farms: prevalence, risk factors and molecular characteristics. J. Antimicrob. Chemother. 69, 2669–2675. https://doi.org/10.1093/jac/dku178.
Huijbers, P.M.C., Larsson, D.G.J., Flach, C.F., 2020. Surveillance of antibiotic resistant *Escherichia coli* in human populations through urban wastewater in ten European countries. Environ. Pollut. 261, 114200. https://doi.org/10.1016/j.envpol.2020.114200.
Hussain, H.I., Iqbal, Z., Seleem, M.N., Huang, D., Sattar, A., Hao, H., Yuan, Z., 2017. Virulence and transcriptome profile of multidrug-resistant *Escherichia coli* from chicken. Sci. Rep. 7, 1–15. https://doi.org/10.1038/s41598-017-07798-1.
Huttner, B., Harbarth, S., 2009. "Antibiotics are not automatic anymore" the French

national campaign to cut antibiotic overuse. PLoS Med. 6, e1000080 https://doi.org/10.1371/journal.pmed.1000080.

Jechalke, S., Schreiter, S., Wolters, B., Dealtry, S., Heuer, H., Smalla, K., 2014. Widespread dissemination of class 1 integron components in soils and related ecosystems as revealed by cultivation-independent analysis. Front. Microbiol. 4, 420. https://doi.org/10.3389/fmicb.2013.00420.

Ji, X., Zheng, B., Berglund, B., Zou, H., Sun, Q., Chi, X., Ottoson, J., Li, X., Lundborg, C.S., Nilsson, L.E., 2019. Dissemination of extended-spectrum β-lactamase-producing *Escherichia coli* carrying *mcr-1* among multiple environmental sources in rural China and associated risk to human health. Environ. Pollut. 251, 619–627. https://doi.org/10.1016/j.envpol.2019.05.002.

Jiang, H., Cheng, H., Liang, Y., Yu, S., Yu, T., Fang, J., Zhu, C., 2019. Diverse mobile genetic elements and conjugal transferability of sulfonamide resistance genes (*sul1, sul2*, and *sul3*) in *Escherichia coli* isolates from *Penaeus vannamei* and pork from large markets in Zhejiang, China. Front. Microbiol. 10, 1787. https://doi.org/10.3389/fmicb.2019.01787.

Jones-Dias, D., Manageiro, V., Ferreira, E., Barreiro, P., Vieira, L., Moura, I.B., Caniça, M., 2016. Architecture of class 1, 2, and 3 integrons from gram negative bacteria recovered among fruits and vegetables. Front. Microbiol. 7, 1400. https://doi.org/10.3389/fmicb.2016.01400.

Kar, D., Bandyopadhyay, Samiran, Bhattacharyya, D., Samanta, I., Mahanti, A., Nanda, P.K., Mondal, B., Dandapat, P., Das, A.K., Dutta, T.K., Bandyopadhyay, Subhasish, Singh, R.K., 2015. Molecular and phylogenetic characterization of multidrug resistant extended spectrum beta-lactamase producing *Escherichia coli* isolated from poultry and cattle in Odisha, India. Infect. Genet. Evol. 29, 82–90. https://doi.org/10.1016/j.meegid.2014.11.003.

Kargar, M., Mohammadalipour, Z., Doosti, A., Lorzadeh, S., Japoni-Nejad, A., 2014. High prevalence of class 1 to 3 integrons among multidrug-resistant diarrheagenic *Escherichia coli* in southwest of Iran. Osong Public Health Res. Perspect 5, 193–198. https://doi.org/10.1016/j.phrp.2014.06.003.

Kaushik, M., Kumar, S., Kapoor, R.K., Virdi, J.S., Gulati, P., 2018. Integrons in *Enterobacteriaceae*: diversity, distribution and epidemiology. Int. J. Antimicrob. Agents 51, 167–176. https://doi.org/10.1016/j.ijantimicag.2017.10.004.

Kayange, N., Kamugisha, E., Mwizamholya, D.L., Jeremiah, S., Mshana, S.E., 2010. Predictors of positive blood culture and deaths among neonates with suspected neonatal sepsis in a tertiary hospital, Mwanza- Tanzania. BMC Pediatr. 10, 39. https://doi.org/10.1186/1471-2431-10-39.

Klein, E.Y., Van Boeckel, T.P., Martinez, E.M., Pant, S., Gandra, S., Levin, S.A., et al., 2018. Global increase and geographic convergence in antibiotic consumption between 2000 and 2015. Proc. Natl. Acad. Sci. Unit. States Am. 115 (15), E3463–E3470. https://doi.org/10.1073/pnas.1717295115.

Koczura, R., Mokracka, J., Jabłońska, L., Gozdecka, E., Kubek, M., Kaznowski, A., 2012. Antimicrobial resistance of integron-harboring *Escherichia coli* isolates from clinical samples, wastewater treatment plant and river water. Sci. Total Environ. 414, 680–685. https://doi.org/10.1016/j.scitotenv.2011.10.036.

Koczura, R., Mokracka, J., Taraszewska, A., Łopacinska, N., 2016. Abundance of class 1 integron-integrase and sulfonamide resistance genes in river water and sediment is affected by anthropogenic pressure and environmental factors. Microb. Ecol. 72, 909–916. https://doi.org/10.1007/s00248-016-0843-4.

Kogovšek, P., Ambrožič-Avguštin, J., Dovč, A., Dreo, T., Hristov, H., Krapež, U., Ravnikar, M., Slavec, B., Lotrič, M., Žel, J., Zorman Rojs, O., 2019. Loop-mediated isothermal amplification: rapid molecular detection of virulence genes associated with avian pathogenic *Escherichia coli* in poultry. Poultry Sci. 98, 1500–1510. https://doi.org/10.3382/ps/pey516.

Kotlarska, E., Łuczkiewicz, A., Pisowacka, M., Burzyński, A., 2015. Antibiotic resistance and prevalence of class 1 and 2 integrons in *Escherichia coli* isolated from two wastewater treatment plants, and their receiving waters (Gulf of Gdansk, Baltic Sea, Poland). Environ. Sci. Pollut. Res. 22 https://doi.org/10.1007/s11356-014-3474-7, 2018–2030.

Kumar, V., Gupta, J., Meena, H.R., 2019. Assessment of awareness about antibiotic resistance and practices followed by veterinarians for judicious prescription of antibiotics: an exploratory study in eastern Haryana region of India. Trop. Anim. Health Prod. 51, 677–687. https://doi.org/10.1007/s11250-018-1742-0.

Lallement, C., Pasternak, C., Ploy, M.-C., Jové, T., 2018. The Role of IS*CR1*-Borne P_{OUT} Promoters in the Expression of Antibiotic Resistance Genes. Front. Microbiol. 9, 2579. https://doi.org/10.3389/fmicb.2018.02579.

Larsson, D.G.J., Andremont, A., Bengtsson-Palme, J., Brandt, K.K., de Roda Husman, A.M., Fagerstedt, P., Fick, J., Flach, C.-F., Gaze, W.H., Kuroda, M., Kvint, K., Laxminarayan, R., Manaia, C.M., Nielsen, K.M., Plant, L., Ploy, M.-C., Segovia, C., Simonet, P., Smalla, K., Snape, J., Topp, E., van Hengel, A.J., Verner-Jeffreys, D.W., Virta, M.P.J., Wellington, E.M., Wernersson, A.-S., 2018. Critical knowledge gaps and research needs related to the environmental dimensions of antibiotic resistance. Environ. Int. 117, 132–138. https://doi.org/10.1016/j.envint.2018.04.041.

Laupland, K.B, Church, D.L, Vidakovich, J., Mucenski, M., Pitout, J.D., 2008. Community-onset extended-spectrum β-lactamase (ESBL) producing *Escherichia coli*: Importance of international travel. J. Infect. 57, 441–448. https://doi.org/10.1016/j.jinf.2008.09.034.

Laxminarayan, R., Duse, A., Wattal, C., Zaidi, A.K.M., Wertheim, H.F.L., Sumpradit, N., Vlieghe, E., Hara, G.L., Gould, I.M., Goossens, H., Greko, C., So, A.D., Bigdeli, M., Tomson, G., Woodhouse, W., Ombaka, E., Peralta, A.Q., Qamar, F.N., Mir, F., Kariuki, S., Bhutta, Z.A., Coates, A., Bergstrom, R., Wright, G.D., Brown, E.D., Cars, O., 2013. Antibiotic resistance—the need for global solutions. Lancet Infect. Dis. 13, 1057–1098. https://doi.org/10.1016/S1473-3099(13)70318-9.

Laxminarayan, R., Van Boeckel, T., Frost, I., Kariuki, S., Khan, E.A., Limmathurotsakul, D., Larsson, D.G., Levy-Hara, G., Mendelson, M., Outterson, K., Peacock, S.J., Zhu, Y., 2020. The lancet infectious diseases commission on antimicrobial resistance: 6 years later. Lancet Infect. Dis. 20, e51–e60. https://doi.org/10.1016/S1473-3099(20)30003-7.

Le-Vo, H.N., Tran, P.T.-B., Le, L., Matsumoto, Y., Motooka, D., Nakamura, S., Jones, J.W., Iida, T., Cao, V., 2019. Complex class 1 integron in a clinical *Escherichia coli* strain from Vietnam carrying both *mcr-1* and bla_{NDM-1}. Front. Microbiol. 10, 2472. https://doi.org/10.3389/fmicb.2019.02472.

Lee, K., Kim, D.-W., Lee, D.-H., Kim, Y.-S., Bu, J.-H., Cha, J.-H., Thawng, C.N., Hwang, E.-M., Seong, H.J., Sul, W.J., Wellington, E.M.H., Quince, C., Cha, C.-J., 2020. Mobile resistome of human gut and pathogen drives anthropogenic bloom of antibiotic resistance. Microbiome 8, 2. https://doi.org/10.1186/s40168-019-0774-7.

Lekunberri, I., Villagrasa, M., Balcázar, J.L., Borrego, C.M., 2017. Contribution of bacteriophage and plasmid DNA to the mobilization of antibiotic resistance genes in a river receiving treated wastewater discharges. Sci. Total Environ. 601–602, 206–209. https://doi.org/10.1016/j.scitotenv.2017.05.174.

Lévesque, C., Brassard, S., Lapointe, J., Roy, P.H., 1994. Diversity and relative strength of tandem promoters for the antibiotic-resistance genes of several integron. Gene 142, 49–54. https://doi.org/10.1016/0378-1119(94)90353-0.

Li, J., 2017. Current status and prospects for in-feed antibiotics in the different stages of pork production - a review. Asian-Australas. J. Anim. Sci. 30, 1667–1673. https://doi.org/10.5713/ajas.17.0418.

Li, M., Ray, P., Knowlton, K.F., Pruden, A., Xia, K., Teets, C., Du, P., 2020a. Fate of pirlimycin and antibiotic resistance genes in dairy manure slurries in response to temperature and pH adjustment. Sci. Total Environ. 710, 136310 https://doi.org/10.1016/j.scitotenv.2019.136310.

Li, M., Ray, P., Teets, C., Pruden, A., Xia, K., Knowlton, K.F., 2020b. Short communication: increasing temperature and pH can facilitate reductions of cephapirin and antibiotic resistance genes in dairy manure slurries. J. Dairy Sci. https://doi.org/10.3168/jds.2019-17453.

Liao, X., Liu, B., Yang, Q., Sun, J., Li, L., Fang, L., Liu, Y., 2013. Comparison of plasmids coharboring 16s rRNA methylase and extended-spectrum β-lactamase genes among *Escherichia coli* isolates from pets and poultry. J. Food Protect. 76, 2018–2023. https://doi.org/10.4315/0362-028X.JFP-13-200.

Li, S., Yao, Q., Liu, J., Wei, D., Zhou, B., Zhu, P., Cui, X., Jin, J., Liu, X., Wang, G., 2020c. Profiles of antibiotic resistome with animal manure application in black soils of northeast China. J. Hazard Mater. 384, 121216 https://doi.org/10.1016/j.jhazmat.2019.121216.

Liao, K., Bai, Y., Huo, Y., Jian, Z., Hu, W., Zhao, C., Qu, J., 2018. Integrating microbial biomass, composition and function to discern the level of anthropogenic activity in a river ecosystem. Environ. Int. 116, 147–155. https://doi.org/10.1016/j.envint.2018.04.003.

Lienert, J., Bürki, T., Escher, B.I., 2007. Reducing micropollutants with source control: substance flow analysis of 212 pharmaceuticals in faeces and urine. Water Sci. Technol. 56, 87–96. https://doi.org/10.2166/wst.2007.560.

Lin, M., Wu, X., Yan, Q., Ma, Y., Huang, L., Qin, Y., Xu, X., 2016. Incidence of antimicrobial-resistance genes and integrons in antibiotic-resistant bacteria isolated from eels and aquaculture ponds. Dis. Aquat. Org. 120, 115–123. https://doi.org/10.3354/dao03013.

Literák, I., Vanko, R., Dolejská, M., Čížek, A., Karpíšková, R., 2007. Antibiotic resistant *Escherichia coli* and *Salmonella* in Russian rooks (*Corvus frugilegus*) wintering in the Czech Republic. Lett. Appl. Microbiol. 45, 616–621. https://doi.org/10.1111/j.1472-765X.2007.02236.x.

Liu, P., Jia, S., He, X., Zhang, X., Ye, L., 2017. Different impacts of manure and chemical fertilizers on bacterial community structure and antibiotic resistance genes in arable soils. Chemosphere 188, 455–464. https://doi.org/10.1016/j.chemosphere.2017.08.162.

Ma, L., Li, B., Jiang, X.T., Wang, Y.L., Xia, Y., Li, A.D., Zhang, T., 2017. Catalogue of antibiotic resistome and host-tracking in drinking water deciphered by a large scale survey. Microbiome 5, 154. https://doi.org/10.1186/s40168-017-0369-0.

Maamar, E., Hammami, S., Alonso, C.A., Dakhli, N., Abbassi, M.S., Ferjani, S., Hamzaoui, Z., Saidani, M., Torres, C., Boutiba-Ben Boubaker, I., 2016. High prevalence of extended-spectrum and plasmidic AmpC beta-lactamase-producing *Escherichia coli* from poultry in Tunisia. Int. J. Food Microbiol. 231, 69–75. https://doi.org/10.1016/j.ijfoodmicro.2016.05.001.

Marathe, N.P., Janzon, A., Kotsakis, S.D., Flach, C.-F., Razavi, M., Berglund, F., Kristiansson, E., Larsson, D.G.J., 2018. Functional metagenomics reveals a novel carbapenem-hydrolyzing mobile beta-lactamase from Indian river sediments contaminated with antibiotic production waste. Environ. Int. 112, 279–286. https://doi.org/10.1016/j.envint.2017.12.036.

Marchant, M., Moreno, M.A., 2013. Dynamics and diversity of *Escherichia coli* in animals and system management of the manure on a commercial farrow-to-finish pig farm. Appl. Environ. Microbiol. 79, 853–859. https://doi.org/10.1128/AEM.02866-12.

Marshall, B.M., Levy, S.B., 2011. Food animals and antimicrobials: impacts on human health. Clin. Microbiol. Rev. 24, 718–733. https://doi.org/10.1128/CMR.00002-11.

Martinez, J.L., 2009. The role of natural environments in the evolution of resistance traits in pathogenic bacteria. P. ROY. SOC. B-BIOL. SCI. 276, 2521–2530. https://doi.org/10.1098/rspb.2009.0320.

Martinez, J.L., Baquero, F., 2000. Mutation frequencies and antibiotic resistance. Antimicrob. Agents Chemother. 44, 1771–1777. https://doi.org/10.1128/aac.44.7.1771-1777.2000.

Martini, M.C., Quiroga, M.P., Pistorio, M., Lagares, A., Centrón, D., Del Papa, M.F.,

2018. Novel environmental class 1 integrons and cassette arrays recovered from an on-farm bio-purification plant. FEMS Microbiol. Ecol. 94 https://doi.org/10.1093/femsec/fix190 fix190.
Mazel, D., 2006. Integrons: agents of bacterial evolution. Nat. Rev. Microbiol. 4, 608–620. https://doi.org/10.1038/nrmicro1462.
McDougall, F., Boardman, W., Gillings, M., Power, M., 2019. Bats as reservoirs of antibiotic resistance determinants: a survey of class 1 integrons in Grey-headed Flying Foxes (*Pteropus poliocephalus*). Infect. Genet. Evol. 70, 107–113. https://doi.org/10.1016/j.meegid.2019.02.022.
Mehdi, Y., Létourneau-Montminy, M.P., Gaucher, M.L., Chorfi, Y., Suresh, G., Rouissi, T., Brar, S.K., Côté, C., Ramirez, A.A., Godbout, S., 2018. Use of antibiotics in broiler production: global impacts and alternatives. Anim. Nutr. 4, 170–178. https://doi.org/10.1016/j.aninu.2018.03.002.
Messaili, C., Messai, Y., Bakour, R., 2019. Virulence gene profiles, antimicrobial resistance and phylogenetic groups of fecal *Escherichia coli* strains isolated from broiler chickens in Algeria. Vet. Ital. 55, 35–46. https://doi.org/10.12834/VetIt.799.3865.2.
Messier, N., Roy, P.H., 2001. Integron integrases possess a unique additional domain necessary for activity. J. Bacteriol. 183, 6699–6706. https://doi.org/10.1128/JB.183.22.6699-6706.2001.
Moawad, A.A., Hotzel, H., Awad, O., Tomaso, H., Neubauer, H., Hafez, H.M., El-Adawy, H., 2017. Occurrence of *Salmonella enterica* and *Escherichia coli* in raw chicken and beef meat in northern Egypt and dissemination of their antibiotic resistance markers. Gut Pathog. 9, 57. https://doi.org/10.1186/s13099-017-0206-9.
Navab-Daneshmand, T., Friedrich, M.N.D., Gächter, M., Montealegre, M.C., Mlambo, L.S., Nhiwatiwa, T., Mosler, H.-J., Julian, T.R., 2018. *Escherichia coli* contamination across multiple environmental compartments (soil, hands, drinking water, and handwashing water) in urban harare: correlations and risk factors. Am. J. Trop. Med. Hyg. 98, 803–813. https://doi.org/10.4269/ajtmh.17-0521.
Navarro-Gonzalez, N., Porrero, M.C., Mentaberre, G., Serrano, E., Mateos, A., Domínguez, L., Lavín, S., 2013. Antimicrobial resistance in indicator *Escherichia coli* isolates from free-ranging livestock and sympatric wild ungulates in a natural environment (northeastern Spain). Appl. Environ. Microbiol. 79, 6184–6186. https://doi.org/10.1128/AEM.01745-13.
Nelson, J.M., Chiller, T.M., Powers, J.H., Angulo, F.J., 2007. Fluoroquinolone-resistant *Campylobacter* species and the withdrawal of fluoroquinolones from use in poultry: a public health success story. Clin. Infect. Dis. 44, 977–980. https://doi.org/10.1086/512369.
Nnadozie, C.F., Odume, O.N., 2019. Freshwater environments as reservoirs of antibiotic resistant bacteria and their role in the dissemination of antibiotic resistance genes. Environ. Pollut. 254, 113067. https://doi.org/10.1016/j.envpol.2019.113067.
Nõlvak, H., Truu, M., Kanger, K., Tampere, M., Espenberg, M., Loit, E., Raave, H., Truu, J., 2016. Inorganic and organic fertilizers impact the abundance and proportion of antibiotic resistance and integron-integrase genes in agricultural grassland soil. Sci. Total Environ. 562, 678–689. https://doi.org/10.1016/j.scitotenv.2016.04.035.
OECD, 2020. Meat Consumption (Indicator). https://doi.org/10.1787/fa290fd0-en. (Accessed 26 May 2020).
Ojo, O.E., Schwarz, S., Michael, G.B., 2016. Detection and characterization of extended-spectrum β-lactamase-producing *Escherichia coli* from chicken production chains in Nigeria. Vet. Microbiol. 194, 62–68. https://doi.org/10.1016/j.vetmic.2016.04.022.
Pal, C., Asiani, K., Arya, S., Rensing, C., Stekel, D.J., Larsson, D.G.J., Hobman, J.L., 2017. Chapter seven - metal resistance and its association with antibiotic resistance. In: Poole, R.K. (Ed.), Advances in Microbial Physiology, vol. 70. Academic Press, pp. 261–313. https://doi.org/10.1016/bs.ampbs.2017.02.001.
Paraoan, C.E.M., Rivera, W.L., Vital, P.G., 2017. Detection of Class I and II integrons for the assessment of antibiotic and multidrug resistance among *Escherichia coli* isolates from agricultural irrigation waters in Bulacan, Philippines. J. Environ. Sci. Health. 52, 306–313. https://doi.org/10.1080/03601234.2017.1281647.
Partridge, S.R., Recchia, G.D., Scaramuzzi, C., Collis, C.M., Stokes, H.W., Hall, R.M., 2000. Definition of the *attI1* site of class 1 integrons. Microbiology 146, 2855–2864. https://doi.org/10.1099/00221287-146-11-2855.
Partridge, S.R., Tsafnat, G., Coiera, E., Iredell, J.R., 2009. Gene cassettes and cassette arrays in mobile resistance integrons. FEMS Microbiol. Rev. 33, 757–784. https://doi.org/10.1111/j.1574-6976.2009.00175.x.
Park, J., Kim, Y.J., Binn-Kim, Seo, K.H., 2018b. Spread of multidrug-resistant *Escherichia coli* harboring integron via swine farm waste water treatment plant. Ecotoxicol. Environ. Saf. 149, 36–42. https://doi.org/10.1016/j.ecoenv.2017.10.071.
Park, J., Kim, J.S., Kim, S., Shin, E., Oh, K.H., Kim, Y., Kim, C.H., Hwang, M.A., Jin, C.M., Na, K., Lee, J., Cho, E., Kang, B.H., Kwak, H.S., Seong, W.K., Kim, J., 2018a. A waterborne outbreak of multiple diarrhoeagenic *Escherichia coli* infections associated with drinking water at a school camp. Int. J. Infect. Dis. 66, 45–50. https://doi.org/10.1016/j.ijid.2017.09.021.
Partridge, S.R., Kwong, S.M., Firth, N., Jensen, S.O., 2018. Mobile genetic elements associated with antimicrobial resistance. Clin. Microbiol. Rev. 31 (17), e00088 https://doi.org/10.1128/CMR.00088-17.
Pazda, M., Kumirska, J., Stepnowski, P., Mulkiewicz, E., 2019. Antibiotic resistance genes identified in wastewater treatment plant systems – a review. Sci. Total Environ. 697, 134023. https://doi.org/10.1016/j.scitotenv.2019.134023.
Pérez-Etayo, L., Berzosa, M., González, D., Vitas, A.I., 2018. Prevalence of integrons and insertion sequences in ESBL-producing *E. coli* isolated from different sources in navarra, Spain. Int. J. Environ. Res. Publ. Health 15, 2308. https://doi.org/10.3390/ijerph15102308.
Piccirillo, A., Giovanardi, D., Dotto, G., Grilli, G., Montesissa, C., Boldrin, C., Salata, C., Giacomelli, M., 2014. Antimicrobial resistance and class 1 and 2 integrons in *Escherichia coli* from meat turkeys in Northern Italy. Avian Pathol. J. 43, 396–405. https://doi.org/10.1080/03079457.2014.943690.
Ponce-Rivas, E., Muñoz-Márquez, M.E., Khan, A.A., 2012. Identification and molecular characterization of class 1 integrons in multiresistant *Escherichia coli* isolates from poultry litter. Appl. Environ. Microbiol. 78, 5444–5447. https://doi.org/10.1128/AEM.00660-12.
Poudel, A., Hathcock, T., Butaye, P., Kang, Y., Price, S., Macklin, K., Walz, P., Cattley, R., Kalalah, A., Adekanmbi, F., Wang, C., 2019. Multidrug-resistant *Escherichia coli, Klebsiella pneumoniae* and *Staphylococcus spp*. in houseflies and blowflies from farms and their environmental settings. Int. J. Environ. Res. Publ. Health 16, 3583. https://doi.org/10.3390/ijerph16193583.
Pouwels, K.B., Muller-Pebody, B., Smieszek, T., Hopkins, S., Robotham, J.V., 2019. Selection and co-selection of antibiotic resistances among *Escherichia coli* by antibiotic use in primary care: an ecological analysis. PloS One 14, e0218134. https://doi.org/10.1371/journal.pone.0218134.
Power, M.L., Samuel, A., Smith, J.J., Stark, J.S., Gillings, M.R., Gordon, D.M., 2016. *Escherichia coli* out in the cold: dissemination of human-derived bacteria into the Antarctic microbiome. Environ. Pollut. 215, 58–65. https://doi.org/10.1016/j.envpol.2016.04.013.
Probert, W.S., Miller, G.M., Ledin, K.E., 2017. Contaminated stream water as source for *Escherichia coli* O157 illness in children. Emerg. Infect. Dis. 23, 1216–1218. https://doi.org/10.3201/eid2307.170226.
Projahn, M., Daehre, K., Roesler, U., Friese, A., 2017. Extended-spectrum-Beta-lactamase- and plasmid-encoded cephamycinase-producing enterobacteria in the broiler hatchery as a potential mode of pseudo-vertical transmission. Appl. Environ. Microbiol. 83 (16), e02364 https://doi.org/10.1128/AEM.02364-16.
Pruden, A., Pei, R., Storteboom, H., Carlson, K.H., 2006. Antibiotic resistance genes as emerging Contaminants: studies in northern Colorado. Environ. Sci. Technol. 40, 7445–7450. https://doi.org/10.1021/es060413l.
Qiao, M., Ying, G., Singer, A.C., Zhu, Y., 2018. Review of antibiotic resistance in China and its environment. Environ. Int. 110, 160–172. https://doi.org/10.1016/j.envint.2017.10.016.
Qiu, J., Zhao, T., Liu, Q., He, J., He, D., Wu, G., Li, Y., Jiang, C., Xu, Z., 2016. Residual veterinary antibiotics in pig excreta after oral administration of sulfonamides. Environ. Geochem. Health 38, 549–556. https://doi.org/10.1007/s10653-015-9740-x.
Quiroga, M.P., Andres, P., Petroni, A., Bistué, A.J.C.S., Guerriero, L., Vargas, L.J., Zorreguieta, A., Tokumoto, M., Quiroga, C., Tolmasky, M.E., Galas, M., Centrón, D., 2007. Complex class 1 integrons with diverse variable regions, including *aac*(6′)*-Ib-cr*, and a novel allele, *qnrB10*, associated with IS*CR1* in clinical enterobacterial isolates from Argentina. Antimicrob. Agents Chemother. 51, 4466–4470. https://doi.org/10.1128/AAC.00726-07.
Radhouani, H., Silva, N., Poeta, P., Torres, C., Correia, S., Igrejas, G., 2014. Potential impact of antimicrobial resistance in wildlife, environment and human health. Front. Microbiol. 5, 23. https://doi.org/10.3389/fmicb.2014.00023.
Rahman, S., Ali, T., Muhammad, N., Umer, T., Saddique, Ahmad, S., Ayaz, S., Han, B., 2018. Characterization and mechanism of dissemination of extended spectrum beta lactamase producers *Escherichia Coli* in food producing animals in Pakistan and China. IEEE, pp. 203–214. https://doi.org/10.1109/IBCAST.2018.8312224, 2018 15th International Bhurban Conference on Applied Sciences and Technology (IBCAST).
Ramadan, H.H., Jackson, C.R., Taha, S.A., Moawad, A.A., Barrett, J.B., Woodley, T.A., 2018. Contribution of healthy chickens to antimicrobial-resistant *Escherichia coli* associated with human extraintestinal infections in Egypt. Vector Borne Zoonotic Dis. 18, 408–416. https://doi.org/10.1089/vbz.2017.2237.
Rehman, M.U., Zhang, H., Huang, S., Iqbal, M.K., Mehmood, K., Luo, H., Li, J., 2017. Characteristics of integrons and associated gene cassettes in antibiotic-resistant *Escherichia coli* isolated from free-ranging food animals in China. J. Food Sci. 82, 1902–1907. https://doi.org/10.1111/1750-3841.13795.
Resistance, A., 2014. Tackling a Crisis for the Health and Wealth of Nations, vol. 2014. Chaired by Jim O'Neill. The Review on Antimicrobial Resistance.
Robinson, T.P., Bu, D.P., Carrique-Mas, J., Fèvre, E.M., Gilbert, M., Grace, D., Hay, S.I., Jiwakanon, J., Kakkar, M., Kariuki, S., Laxminarayan, R., Lubroth, J., Magnusson, U., Thi Ngoc, P., Van Boeckel, T.P., Woolhouse, M.E.J., 2016. Antibiotic resistance is the quintessential One Health issue. Trans. R. Soc. Trop. Med. Hyg. 110, 377–380. https://doi.org/10.1093/trstmh/trw048.
Roth, N., Käsbohrer, A., Mayrhofer, S., Zitz, U., Hofacre, C., Domig, K.J., 2019. The application of antibiotics in broiler production and the resulting antibiotic resistance in *Escherichia coli*: a global overview. Poultry Sci. 98, 1791–1804. https://doi.org/10.3382/ps/pey539.
Rui, Y., Lu, W., Li, S., Cheng, C., Sun, J., Yang, Q., 2018. Integrons and insertion sequence common region 1 (IS*CR1*) of carbapenem-non-susceptible Gram-negative bacilli in fecal specimens from 5000 patients in southern China. Int. J. Antimicrob. Agents 52, 571–576. https://doi.org/10.1016/j.ijantimicag.2018.06.015.
Rutgersson, C., Ebmeyer, S., Lassen, S.B., Karkman, A., Fick, J., Kristiansson, E., Brandt, K.K., Flach, C.-F., Larsson, D.G.J., 2020. Long-term application of Swedish sewage sludge on farmland does not cause clear changes in the soil bacterial resistome. Environ. Int. 137, 105339. https://doi.org/10.1016/j.envint.2019.105339.

Sabuncu, E., David, J., Bernède-Bauduin, C., Pépin, S., Leroy, M., Boëlle, P.Y., Watier, L., Guillemot, D., 2009. Significant reduction of antibiotic use in the community after a nationwide campaign in France, 2002-2007. PLoS Med. 6, e1000084 https://doi.org/10.1371/journal.pmed.1000084.
Sacristán, C., Esperón, F., Herrera-León, S., Iglesias, I., Neves, E., Nogal, V., Muñoz, M.J., Torre, A. de la, 2014. Virulence genes, antibiotic resistance and integrons in *Escherichia coli* strains isolated from synanthropic birds from Spain. Avian Pathol. 43, 172–175. https://doi.org/10.1080/03079457.2014.897683.
Samanta, I., Bandyopadhyay, S., 2020. Chapter 3 - the emergence of antimicrobial-resistant bacteria in livestock, poultry and agriculture. In: Samanta, I., Bandyopadhyay, S. (Eds.), Antimicrobial Resistance in Agriculture. Academic Press, pp. 19–27. https://doi.org/10.1016/B978-0-12-815770-1.00003-1.
Santos, L., Ramos, F., 2018. Antimicrobial resistance in aquaculture: current knowledge and alternatives to tackle the problem. Int. J. Antimicrob. Agents 52, 135–143. https://doi.org/10.1016/j.ijantimicag.2018.03.010.
Santos, C., Caetano, T., Ferreira, S., Ramalheira, E., Mendo, S., 2011. A novel complex class 1 integron found in a *Klebsiella pneumoniae* isolate from Portugal. Clin. Microbiol. Infect. 17, 1036–1039. https://doi.org/10.1111/j.1469-0691.2010.03416.x.
Santos, T., Silva, N., Igrejas, G., Rodrigues, P., Micael, J., Rodrigues, T., Resendes, R., Gonçalves, A., Marinho, C., Gonçalves, D., Cunha, R., Poeta, P., 2013. Dissemination of antibiotic resistant *Enterococcus* spp. and *Escherichia coli* from wild birds of Azores Archipelago. Anaerobe 24, 25–31. https://doi.org/10.1016/j.anaerobe.2013.09.004.
Sarmah, A.K., Meyer, M.T., Boxall, A.B.A., 2006. A global perspective on the use, sales, exposure pathways, occurrence, fate and effects of veterinary antibiotics (VAs) in the environment. Chemosphere 65, 725–759. https://doi.org/10.1016/j.chemosphere.2006.03.026.
Schwaiger, K., Bauer, J., Hölzel, C.S., 2013. Selection and persistence of antimicrobial-resistant *Escherichia coli* including extended-spectrum β-lactamase producers in different poultry flocks on one chicken farm. Microb. Drug Resist. 19, 498–506. https://doi.org/10.1089/mdr.2012.0257.
Seiffert, S.N., Hilty, M., Kronenberg, A., Droz, S., Perreten, V., Endimiani, A., 2013. Extended-spectrum cephalosporin-resistant *Escherichia coli* in community, specialized outpatient clinic and hospital settings in Switzerland. J. Antimicrob. Chemother. 68, 2249–2254. https://doi.org/10.1093/jac/dkt208.
Shlaes, D.M., Gerding, D.N., John, J.F., Craig, W.A., Bornstein, D.L., Duncan, R.A., Eckman, M.R., Farrer, W.E., Greene, W.H., Lorian, V., Levy, S., McGowan, J.E., Paul, S.M., Ruskin, J., Tenover, F.C., Watanakunakorn, C., 1997. Society for healthcare epidemiology of America and infectious diseases society of America joint committee on the prevention of antimicrobial resistance guidelines for the prevention of antimicrobial resistance in hospitals. Infect. Cont. Hosp. Ep. 18, 275–291. https://doi.org/10.2307/30141215.
Sidhu, J.P.S., Skelly, E., Hodgers, L., Ahmed, W., Li, Y., Toze, S., 2014. Prevalence of *enterococcus* species and their virulence genes in fresh water prior to and after storm events. Environ. Sci. Technol. 48, 2979–2988. https://doi.org/10.1021/es4050083.
Sidhu, J.P.S., Jagals, P., Smith, A., Toze, S., 2017. Comparative prevalence of *Escherichia coli* carrying virulence genes and class 1 and 2 integrons in sub-tropical and cool temperate freshwater. Environ. Sci. Pollut. Res. 24, 18263–18272. https://doi.org/10.1007/s11356-017-9497-0.
Silbergeld, E.K., Graham, J., Price, L.B., 2008. Industrial food animal production, antimicrobial resistance, and human health. Annu. Rev. Publ. Health 29, 151–169. https://doi.org/10.1146/annurev.publhealth.29.020907.090904.
Silva, N., Costa, L., Gonçalves, A., Sousa, M., Radhouani, H., Brito, F., Igrejas, G., Poeta, P., 2012. Genetic characterisation of extended-spectrum β-lactamases in *Escherichia coli* isolated from retail chicken products including CTX-M-9 containing isolates: a food safety risk factor. Br. Poultry Sci. 53, 747–755. https://doi.org/10.1080/00071668.2012.740554.
Silva, V., Carvalho, I., Igrejas, G., Poeta, P., 2017. Soil antibiotics and transfer of antibiotic resistance genes affecting wildlife. Antibiotics and Antibiotics Resistance Genes in Soils. Springer, Cham, pp. 313–325. https://doi.org/10.1007/978-3-319-66260-2_22.
Soare, E., Chiurciu, I.-A., 2017. STUDY ON the pork market worldwide scientific papers: management. In: Economic Engineering in Agriculture & Rural Development, vol. 17, pp. 321–326.
Solà-Ginés, M., González-López, J.J., Cameron-Veas, K., Piedra-Carrasco, N., Cerdà-Cuéllar, M., Migura-Garcia, L., 2015. Houseflies (*Musca domestica*) as vectors for extended-spectrum β-lactamase-producing *Escherichia coli* on Spanish broiler farms. Appl. Environ. Microbiol. 81, 3604–3611. https://doi.org/10.1128/AEM.04252-14.
Song, J., Jang, S., Lee, J.J., Lee, J.H., Bae, I.K., Jeong, B.C., Cha, S.-S., Lee, J.-H., Hong, S.-K., Lee, S.H., 2010. Association of the bla_{CMY-10} gene with a novel complex class 1 integron carrying an IS*CR1* element in clinical isolates from Korea. Clin. Microbiol. Infect. 16, 1013–1017. https://doi.org/10.1111/j.1469-0691.2009.03002.x.
Soufi, L., Sáenz, Y., Vinué, L., Abbassi, M.S., Ruiz, E., Zarazaga, M., Ben Hassen, A., Hammami, S., Torres, C., 2011. *Escherichia coli* of poultry food origin as reservoir of sulphonamide resistance genes and integrons. Int. J. Food Microbiol. 144, 497–502. https://doi.org/10.1016/j.ijfoodmicro.2010.11.008.
Speksnijder, D.C., Wagenaar, J.A., 2018. Reducing antimicrobial use in farm animals: how to support behavioral change of veterinarians and farmers. Anim. Front. 8, 4–9. https://doi.org/10.1093/af/vfy006.
Stark, J.S., Corbett, P.A., Dunshea, G., Johnstone, G., King, C., Mondon, J.A., Power, M.L., Samuel, A., Snape, I., Riddle, M., 2016. The environmental impact of sewage and wastewater outfalls in Antarctica: an example from Davis station, East Antarctica. Water Res. 105, 602–614. https://doi.org/10.1016/j.watres.2016.09.026.
Stokes, H.W., O'Gorman, D.B., Recchia, G.D., Parsekhian, M., Hall, R.M., 1997. Structure and function of 59-base element recombination sites associated with mobile gene cassettes. Mol. Microbiol. 26, 731–745. https://doi.org/10.1046/j.1365-2958.1997.6091980.x.
Stromberg, Z.R., Johnson, J.R., Fairbrother, J.M., Kilbourne, J., Van Goor, A., Curtiss, R., Mellata, M., 2017. Evaluation of *Escherichia coli* isolates from healthy chickens to determine their potential risk to poultry and human health. PloS One 12. https://doi.org/10.1371/journal.pone.0180599.
Su, H., Ying, G., Tao, R., Zhang, R., Zhao, J., Liu, Y., 2012. Class 1 and 2 integrons, *sul* resistance genes and antibiotic resistance in *Escherichia coli* isolated from Dongjiang River, South China. Environ. Pollut. 169, 42–49. https://doi.org/10.1016/j.envpol.2012.05.007.
Subbiah, M., Shah, D.H., Besser, T.E., Ullman, J.L., Call, D.R., 2012. Urine from treated cattle drives selection for cephalosporin resistant *Escherichia coli* in soil. PloS One 7, e48919. https://doi.org/10.1371/journal.pone.0048919.
Sui, Q., Chen, Y., Yu, D., Wang, T., Hai, Y., Zhang, J., Chen, M., Wei, Y., 2019. Fates of intracellular and extracellular antibiotic resistance genes and microbial community structures in typical swine wastewater treatment processes. Environ. Int. 133, 105183. https://doi.org/10.1016/j.envint.2019.105183.
Sun, C., Su, Z., Zhou, C., Liu, Y., Yuan, H., Yin, J., Xu, H., 2012. Complex class 1 integron containing $bla_{CTX-M-1}$ genes isolated from *Escherichia coli*: a potentially novel resistant gene-capturing tool kit. Curr. Microbiol. 64, 265–270. https://doi.org/10.1007/s00284-011-0062-7.
Sun, J., Chen, C., Cui, C., Zhang, Y., Liu, X., Cui, Z., Ma, X., Feng, Y., Fang, L., Lian, X., Zhang, R., Tang, Y., Zhang, K., Liu, H., Zhuang, Z., Zhou, S., Lv, J., Du, H., Huang, B., Yu, F., Mathema, B., Kreiswirth, B., Liao, X., Chen, L., Liu, Y., 2019. Plasmid-encoded *tet*(X) genes that confer high-level tigecycline resistance in *Escherichia coli*. Nat. Microbiol. 4, 1457–1464. https://doi.org/10.1038/s41564-019-0496-4.
Sunde, M., Simonsen, G.S., Slettemeås, J.S., Böckerman, I., Norström, M., 2015. Integron, plasmid and host strain characteristics of *Escherichia coli* from humans and food included in the Norwegian antimicrobial resistance monitoring programs. PloS One 10, e0128797. https://doi.org/10.1371/journal.pone.0128797.
Tacconelli, E., Carrara, E., Savoldi, A., Harbarth, S., Mendelson, M., Monnet, D.L., Pulcini, C., Kahlmeter, G., Kluytmans, J., Carmeli, Y., Ouellette, M., Outterson, K., Patel, J., Cavaleri, M., Cox, E.M., Houchens, C.R., Grayson, M.L., Hansen, P., Singh, N., Theuretzbacher, U., Magrini, N., the WHO Pathogens Priority List Working Group, 2018a. Discovery, research, and development of new antibiotics: the WHO priority list of antibiotic-resistant bacteria and tuberculosis. Lancet Infect. Dis. 18, 318–327. https://doi.org/10.1016/S1473-3099(17)30753-3.
Tacconelli, E., Sifakis, F., Harbarth, S., Schrijver, R., van Mourik, M., Voss, A., Sharland, M., Rajendran, N.B., Rodríguez-Baño, J., EPI-Net COMBACTE-MAGNET Group, 2018b. Surveillance for control of antimicrobial resistance. Lancet Infect. Dis. 18, e99–e106. https://doi.org/10.1016/S1473-3099(17)30485-1.
Tajbakhsh, E., Khamesipour, F., Ranjbar, R., Ugwu, I.C., 2015. Prevalence of class 1 and 2 integrons in multi-drug resistant *Escherichia coli* isolated from aquaculture water in Chaharmahal Va Bakhtiari province, Iran. Ann. Clin. Microbiol. Antimicrob. 14, 37. https://doi.org/10.1186/s12941-015-0096-y.
Tewari, R., Mitra, S., Ganaie, F., Das, S., Chakraborty, A., Venugopal, N., Shome, R., Rahman, H., Shome, B.R., 2019. Dissemination and characterisation of *Escherichia coli* producing extended-spectrum β-lactamases, AmpC β-lactamases and metallo-β-lactamases from livestock and poultry in Northeast India: a molecular surveillance approach. J. Glob. Antimicrob. Resist. 17, 209–215. https://doi.org/10.1016/j.jgar.2018.12.025.
Thanner, S., Drissner, D., Walsh, F., 2016. Antimicrobial resistance in agriculture. mBio 7, e02227. https://doi.org/10.1128/mBio.02227-15.
Theobald, S., Etter, E.M.C., Gerber, D., Abolnik, C., 2019. Antimicrobial resistance trends in *Escherichia coli* in South African poultry: 2009–2015. Foodb. Pathog. Dis. 16, 652–660. https://doi.org/10.1089/fpd.2018.2612.
Toleman, M.A., Bennett, P.M., Walsh, T.R., 2006. IS*CR* elements: novel gene-capturing systems of the 21st century? Microbiol. Mol. Biol. Rev. 70, 296–316. https://doi.org/10.1128/MMBR.00048-05.
Torres, R.T., Fernandes, J., Carvalho, J., Cunha, M.V., Caetano, T., Mendo, S., Serrano, E., Fonseca, C., 2020. Wild boar as a reservoir of antimicrobial resistance. Sci. Total Environ. 717, 135001. https://doi.org/10.1016/j.scitotenv.2019.135001.
Tullo, E., Finzi, A., Guarino, M., 2019. Review: environmental impact of livestock farming and Precision Livestock Farming as a mitigation strategy. Sci. Total Environ. 650, 2751–2760. https://doi.org/10.1016/j.scitotenv.2018.10.018.
Unno, T., Han, D., Jang, J., Lee, S.N., Kim, J.H., Ko, G., Kim, B.G., Ahn, J.H., Kanaly, R.A., Sadowsky, M.J., Hur, H.G., 2010. High diversity and abundance of antibiotic-resistant *Escherichia coli* isolated from humans and farm animal hosts in Jeonnam Province, South Korea. Sci. Total Environ. 408, 3499–3506. https://doi.org/10.1016/j.scitotenv.2010.04.046.
Urra, J., Alkorta, I., Garbisu, C., 2019. Potential benefits and risks for soil health derived from the use of organic amendments in agriculture. Agronomy 9, 542. https://doi.org/10.3390/agronomy9090542.
Valentin, L., Sharp, H., Hille, K., Seibt, U., Fischer, J., Pfeifer, Y., Michael, G.B., Nickel, S., Schmiedel, J., Falgenhauer, L., Friese, A., Bauerfeind, R., Roesler, U., Imirzalioglu, C., Chakraborty, T., Helmuth, R., Valenza, G., Werner, G., Schwarz, S., Guerra, B., Appel, B., Kreienbrock, L., Käsbohrer, A., 2014. Subgrouping of ESBL-producing *Escherichia coli* from animal and human sources:

an approach to quantify the distribution of ESBL types between different reservoirs. Int. J. Med. Microbiol 304, 805–816. https://doi.org/10.1016/j.ijmm.2014.07.015.

Van Boeckel, T.P., Brower, C., Gilbert, M., Grenfell, B.T., Levin, S.A., Robinson, T.P., Teillant, A., Laxminarayan, R., 2015. Global trends in antimicrobial use in food animals. Proc. Natl. Acad. Sci. Unit. States Am. 112, 5649–5654. https://doi.org/10.1073/pnas.1503141112.

Velhner, M., Todorović, D., Grego, E., Jovčić, B., Prunić, B., Stojanov, I., Kehrenberg, C., 2018. Fluoroquinolone-resistant and extended-spectrum beta-lactamase producing *Escherichia coli* isolates from free-living wild animals. Vet. Microbiol. 223, 168–172. https://doi.org/10.1016/j.vetmic.2018.08.011.

Wallace, J.S., Garner, E., Pruden, A., Aga, D.S., 2018. Occurrence and transformation of veterinary antibiotics and antibiotic resistance genes in dairy manure treated by advanced anaerobic digestion and conventional treatment methods. Environ. Pollut. 236, 764–772. https://doi.org/10.1016/j.envpol.2018.02.024.

Walsh, T.R., Wu, Y., 2016. China bans colistin as a feed additive for animals. Lancet Infect. Dis. 16, 1102–1103. https://doi.org/10.1016/S1473-3099(16)30329-2.

Wambugu, P., Kiiru, J., Matiru, V., 2018. *Escherichia coli* harbouring resistance genes, virulence genes and integron 1 isolated from athi river in Kenya. Adv. Microbiol. 8, 846–858. https://doi.org/10.4236/aim.2018.811056.

Wang, H., Tang, C., Yang, J., Wang, N., Jiang, F., Xia, Q., He, G., Chen, Y., Jiang, Q., 2018a. Predictors of urinary antibiotics in children of Shanghai and health risk assessment. Environ. Int. 121, 507–514. https://doi.org/10.1016/j.envint.2018.09.032.

Wang, M., Liu, P., Zhou, Q., Tao, W., Sun, Y., Zeng, Z., 2018b. Estimating the contribution of bacteriophage to the dissemination of antibiotic resistance genes in pig feces. Environ. Pollut. 238, 291–298. https://doi.org/10.1016/j.envpol.2018.03.024.

Wang, Y., Xu, C., Zhang, R., Chen, Y., Shen, Y., Hu, F., Liu, D., Lu, J., Guo, Y., Xia, X., Jiang, J., Wang, X., Fu, Y., Yang, L., Wang, J., Li, J., Cai, C., Yin, D., Che, J., Fan, R., Wang, Y., Qing, Y., Li, Y., Liao, K., Chen, H., Zou, M., Liang, L., Tang, J., Shen, Z., Wang, S., Yang, X., Wu, C., Xu, S., Walsh, T.R., Shen, J., 2020. Changes in colistin resistance and *mcr-1* abundance in *Escherichia coli* of animal and human origins following the ban of colistin-positive additives in China: an epidemiological comparative study. Lancet Infect. Dis. https://doi.org/10.1016/S1473-3099(20)30149-3.

Wang, Y., Zhang, R., Li, J., Wu, Z., Yin, W., Schwarz, S., Tyrrell, J.M., Zheng, Y., Wang, S., Shen, Z., Liu, Z., Liu, J., Lei, L., Li, M., Zhang, Q., Wu, C., Zhang, Q., Wu, Y., Walsh, T.R., Shen, J., 2017. Comprehensive resistome analysis reveals the prevalence of NDM and MCR-1 in Chinese poultry production. Nat. Microbiol. 2, 1–7. https://doi.org/10.1038/nmicrobiol.2016.260.

Wernicki, A., Nowaczek, A., Urban-Chmiel, R., 2017. Bacteriophage therapy to combat bacterial infections in poultry. Virol. J. 14, 179. https://doi.org/10.1186/s12985-017-0849-7.

Wolters, B., Widyasari-Mehta, A., Kreuzig, R., Smalla, K., 2016. Contaminations of organic fertilizers with antibiotic residues, resistance genes, and mobile genetic elements mirroring antibiotic use in livestock? Appl. Microbiol. Biotechnol. 100, 9343–9353. https://doi.org/10.1007/s00253-016-7742-y.

Woolhouse, M., Ward, M., van Bunnik, B., Farrar, J., 2015. Antimicrobial resistance in humans, livestock and the wider environment. Philos. Trans. R. Soc. Lond B Biol. Sci. 370, 20140083. https://doi.org/10.1098/rstb.2014.0083.

World Health Organization, 2000. WHO global principles for the containment of antimicrobial resistance in animals intended for food: report of a WHO consultation with the participation of the Food and Agriculture Organization of the United Nations and the Office International des Epizooties, Geneva, Switzerland 5-9 June 2000. World Health Organization, Geneva.

Wright, G.D., 2010. Antibiotic resistance in the environment: a link to the clinic? Curr. Opin. Microbiol. 13, 589–594. https://doi.org/10.1016/j.mib.2010.08.005.

Wu, H., Xia, S., Bu, F., Qi, J., Liu, Y., Xu, H., 2015. Identification of integrons and phylogenetic groups of drug-resistant *Escherichia coli* from broiler carcasses in China. Int. J. Food Microbiol. 211, 51–56. https://doi.org/10.1016/j.ijfoodmicro.2015.07.004.

Wu, J., Huang, Y., Rao, D., Zhang, Y., Yang, K., 2018. Evidence for environmental dissemination of antibiotic resistance mediated by wild birds. Front. Microbiol. 9, 745. https://doi.org/10.3389/fmicb.2018.00745.

Xiong, W., Sun, Y., Zeng, Z., 2018a. Antimicrobial use and antimicrobial resistance in food animals. Environ. Sci. Pollut. Res. 25, 18377–18384. https://doi.org/10.1007/s11356-018-1852-2.

Xiong, L., Sun, Y., Shi, L., Yan, H., 2019. Characterization of antimicrobial resistance genes and class 1 integrase gene in raw meat and aquatic product, fresh vegetable and fruit, and swine manure in southern China. Food Contr. 104, 240–246. https://doi.org/10.1016/j.foodcont.2019.05.004.

Xiong, W., Wang, Y., Sun, Y., Ma, L., Zeng, Q., Jiang, X., Li, A., Zeng, Z., Zhang, T., 2018b. Antibiotic-mediated changes in the fecal microbiome of broiler chickens define the incidence of antibiotic resistance genes. Microbiome 6, 34. https://doi.org/10.1186/s40168-018-0419-2.

Xu, Z., Li, L., Shi, L., Shirtliff, M.E., 2011. Class 1 integron in staphylococci. Mol. Biol. Rep. 38, 5261–5279. https://doi.org/10.1007/s11033-011-0676-7.

Yadav, S., Jha, R., 2019. Strategies to modulate the intestinal microbiota and their effects on nutrient utilization, performance, and health of poultry. J. Anim. Sci. Biotechnol. 10, 1–11. https://doi.org/10.1186/s40104-018-0310-9.

Yang, L., Yang, L., Lü, D., Zhang, W., Ren, S., Liu, Y., Zeng, Z., Jiang, H., 2015. Coprevalance of PMQR and 16S rRNA methylase genes in clinical *Escherichia coli* isolates with high diversity of CTX-M from diseased farmed pigeons. Vet. Microbiol. 178, 238–245. https://doi.org/10.1016/j.vetmic.2015.05.009.

Yang, Y., Shi, W., Lu, S.Y., Liu, J., Liang, H., Yang, Y., Duan, G., Li, Y., Wang, H., Zhang, A., 2018. Prevalence of antibiotic resistance genes in bacteriophage DNA fraction from Funan River water in Sichuan, China. Sci. Total Environ. 626, 835–841. https://doi.org/10.1016/j.scitotenv.2018.01.148.

Yang, Y., Xie, X., Tang, M., Liu, J., Tuo, H., Gu, J., Tang, Y., Lei, C., Wang, H., Zhang, A., 2020. Exploring the profile of antimicrobial resistance genes harboring by bacteriophage in chicken feces. Sci. Total Environ. 700, 134446. https://doi.org/10.1016/j.scitotenv.2019.134446.

Yao, X., Doi, Y., Zeng, L., Lv, L., Liu, J.H., 2016. Carbapenem-resistant and colistin-resistant *Escherichia coli* co-producing NDM-9 and MCR-1. Lancet Infect. Dis. 16, 288–289. https://doi.org/10.1016/S1473-3099(16)00057-8.

Zhang, L., Levy, K., Trueba, G., Cevallos, W., Trostle, J., Foxman, B., Marrs, C.F., Eisenberg, J.N.S., 2015. Effects of selection pressure and genetic association on the relationship between antibiotic resistance and virulence in *Escherichia coli*. Antimicrob. Agents Chemother. 59, 6733–6740. https://doi.org/10.1128/AAC.01094-15.

Zhang, S., Yang, H., Rehman, M.U., Yang, K., Dong, M., Yang, J., Wang, M., Jia, R., Chen, S., Liu, M., Zhu, D., Zhao, X., Yang, Q., Wu, Y., Zhang, L., Liu, Y., Yu, Y., Tian, B., Pan, L., Chen, X., Cheng, A., 2019. Class 1 integrons as predominant carriers in *Escherichia coli* isolates from waterfowls in Hainan, China. Ecotoxicol. Environ. Saf. 183, 109514 https://doi.org/10.1016/j.ecoenv.2019.109514.

Zhao, S., Blickenstaff, K., Bodeis-Jones, S., Gaines, S.A., Tong, E., McDermott, P.F., 2012. Comparison of the prevalences and antimicrobial resistances of *Escherichia coli* isolates from different retail meats in the United States, 2002 to 2008. Appl. Environ. Microbiol. 78, 1701–1707. https://doi.org/10.1128/AEM.07522-11.

Zheng, B., Dong, H., Xu, H., Lv, J., Zhang, J., Jiang, X., Du, Y., Xiao, Y., Li, L., 2016. Coexistence of MCR-1 and NDM-1 in clinical *Escherichia coli* isolates. Clin. Infect. Dis. 63, 1393–1395. https://doi.org/10.1093/cid/ciw553.

Zhou, Z., Feng, W., Han, Y., Zheng, J., Chen, T., Wei, Y., Gillings, M., Zhu, Y., Chen, H., 2018. Prevalence and transmission of antibiotic resistance and microbiota between humans and water environments. Environ. Int. 121, 1155–1161. https://doi.org/10.1016/j.envint.2018.10.032.

Zhu, Y., Zhao, Y., Zhu, D., Gillings, M., Penuelas, J., Ok, Y.S., Capon, A., Banwart, S., 2019. Soil biota, antimicrobial resistance and planetary health. Environ. Int. 131, 105059 https://doi.org/10.1016/j.envint.2019.105059.

Zhou, X., Qiao, M., Su, J., Wang, Y., Cao, Z., Cheng, W., Zhu, Y., 2019b. Turning pig manure into biochar can effectively mitigate antibiotic resistance genes as organic fertilizer. Sci. Total Environ. 649, 902–908. https://doi.org/10.1016/j.scitotenv.2018.08.368.

Zhou, S., Zhu, D., Giles, M., Yang, X., Daniell, T., Neilson, R., Zhu, Y., 2019a. Phyllosphere of staple crops under pig manure fertilization, a reservoir of antibiotic resistance genes. Environ. Pollut. 252, 227–235. https://doi.org/10.1016/j.envpol.2019.05.098.

Zhu, Z., Pan, S., Wei, B., Liu, H., Zhou, Z., Huang, X., Luo, Y., Zhou, L., Zhang, S., Ma, X., Cao, S., Shen, L., Wang, Y., Fu, H., Geng, Y., He, C., Xie, Y., Peng, G., Zhong, Z., 2020. High prevalence of multi-drug resistances and diversity of mobile genetic elements in *Escherichia coli* isolates from captive giant pandas. Ecotoxicol. Environ. Saf. 198, 110681 https://doi.org/10.1016/j.ecoenv.2020.110681.